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PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

TWO-STEP ELECTROLYTIC COLOR ANODIZING

PROCESS VARIANT **2-STEP ELECTROLYTIC**

= CLEAR ANODIZE + AC METAL COLOR IN THE PORES

NO DYE · NO ALLOY-COLOR

ARCHITECTURAL · METAL DEPOSITED IN THE PORES (AC)

CLEAR SULFURIC ANODIZE → AC TIN/NICKEL/COBALT COLORING · CHAMPAGNE→BRONZE→BLACK · LIGHTFAST · AAMA 611 CLASS I

THE COMPLETE TRAINING MANUAL

*A full training course for the brand-new operator — from "wait, the part is the positive electrode?" to a signed certificate of completion. This is the **modern architectural color standard**: first we grow a **clear, ordinary sulfuric (Type II) anodize**, then we move the part to a **second tank of metal salts and color it with AC**, depositing fine metal **deep in the bottom of the pores**. The result is a durable, lightfast champagne-to-bronze-to-black that holds for decades — **more energy-efficient and more consistent than integral color.** Assumes you know nothing. Teaches you everything.*

01 INSP/RACK ▸ 02 CLEAN ▸ 03 ETCH ▸ 04 RINSE ▸ 05 DESMUT ▸ 06 RINSE

07 CLEAR ANODIZE ▸ 08 RINSE ▸ 09 AC COLOR ▸ 10 RINSE ▸ 11 SEAL ▸ 12 RINSE/DRY ▸ 13 INSPECT

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · FOR-DUMMIES STYLE

FIND YOUR WAY

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★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, your customer's spec (AAMA 611, ASTM B580), or your supervisor's instructions. When this manual and your shop's procedure disagree, **follow your shop's official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN THE RIGHT PLACE — HERE THE TRICK IS TWO TANKS: GROW A CLEAR ANODIZE, THEN PLANT DURABLE COLOR METAL IN THE PORES WITH AC

Welcome. You are about to learn how to run — or at least understand — a **two-step electrolytic color anodizing line**, the process the trade also calls **two-step color**, **electrolytic coloring**, or **electrocoloring**. Maybe you started this week; maybe you've racked parts for a year and finally want to know *why*. Either way, you're in the right place.

Here is the first big thing, and it'll feel backwards if you've seen electroplating: **in anodizing, we are not coating your part with a different metal. We are growing a tough, ceramic oxide layer *out of the aluminum itself*.** And to do that, your part is the **anode** — the *positive* electrode — the exact opposite of plating. That's the single biggest *process* idea in this whole book.

Here is the second big idea, and it is the reason this process exists: **two-step color is exactly what its name says — two steps.** First (**Step 1**) we grow a **normal, clear sulfuric (Type II) anodize.** Then (**Step 2**) we move the part to a **second tank of dissolved metal salts** (most often **tin**, sometimes **nickel** or **cobalt**) and run **AC — alternating current.** The AC drives fine particles of that metal **deep into the *bottom* of the pores**, where they **scatter and absorb light** to give a color from **champagne → bronze → black.** Then we seal.

Why has it taken over from integral color? Three reasons: it is **more energy-efficient** (a short, low-energy AC dip, not a thick high-voltage anodize), **far more consistent** across alloys and lots (you *deposit* the color yourself), and gives a **wider tonal range — especially excellent blacks.**

> ★ **REMEMBER**
> Do not read "modern and efficient" as "easy." A two-step line has **everything a Type II line has** — strong sulfuric acid, hot caustic etch, flammable hydrogen, high-current DC, a near-boiling seal — **plus a color tank that puts AC electrical current into an acidic, metal-salt solution.** Tin is comparatively mild, but **nickel and cobalt salts are sensitizers and carcinogen concerns by inhalation,** and the metal waste is regulated. Respect both tanks.

And the third thing: **this manual assumes you know nothing** — every term is defined the first time we use it, in the patient spirit of the *For Dummies* books. *Understand* it; don't just memorize it.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels — from inspect/rack through **clear-anodize (Step 1)**, **AC color (Step 2)**, seal, and final inspect. Reading it in order matters: **the order of an anodizing line is not random** — and neither is this book's.

> 💡 **TIP**
> Read the Fundamentals (Part One) through *before* any station chapter — **especially Chapter 3 (the three coloring routes) and Chapter 4 (how AC coloring works).** The station chapters assume you already grasp the big picture: the clear coat is grown first, and **the color is metal deposited in the pore bottoms — its depth set by how long and how hard you color, not by anodize thickness.**

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

> 💡 **TIP**
> A trick of the trade — what a 20-year veteran would lean over and tell you.

> ★ **REMEMBER**
> A core, load-bearing idea worth locking in.

> ⚠ **HEADS-UP**
> A safety warning — read every one; they protect your skin, eyes, lungs, and health.

> ⚙ **TECHNICAL STUFF**
> Deeper chemistry for the curious. Optional — skip it and you can still run the line.

> ☀ **FUN FACT**
> A bit of trivia to make the science stick (and to give you something to say at lunch). Each station chapter also closes with a **Teach-Back** checklist and a **Sign-Off** box your trainer initials once you've shown the skill safely.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

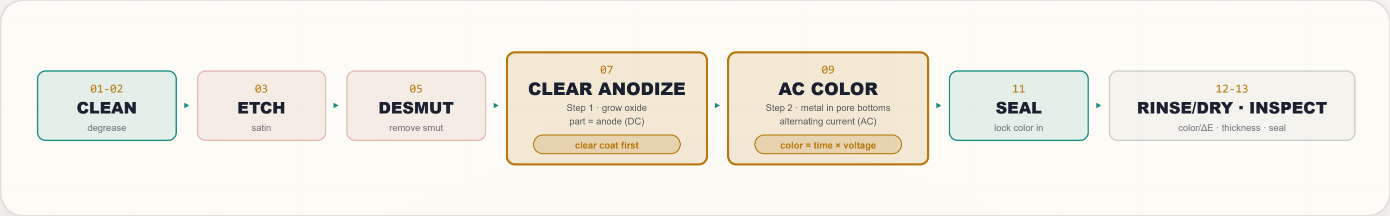
1. **Explain what anodizing is** in plain language — and how it is fundamentally *different from electroplating*: the part is the **anode**, and we **grow an aluminum-oxide layer out of the aluminum itself** instead of depositing a different metal onto it.
2. **State what makes two-step color different:** it is a **two-step process** — (Step 1) a normal **clear sulfuric (Type II) anodize**, then (Step 2) **AC electrolytic coloring** in a metal-salt bath that deposits **tin, nickel, or cobalt metal deep in the bottom of the pores.** The color is durable metal, not dye.
3. **Explain how the color forms** — on the cathodic half-cycle of the AC, metal ions are **reduced and deposited at the pore base**, where they **scatter and absorb light** to give champagne→bronze→black. **Color depth is set by coloring TIME and AC VOLTAGE**, not by coating thickness.
4. **Explain why two-step is *the* modern architectural color** — curtain walls, storefronts, window/door systems — because it is **lightfast and durable like integral color, but more energy-efficient, far more consistent across alloys/lots, and gives superior blacks and bronzes** — and recognize where it sits under **AAMA 611** and **ASTM B580 / architectural Class I (≥18 µm / 0.7 mil).**
5. **Compare the three coloring routes** — **dyeing** (cheap, any color, least lightfast — interior), **integral color** (durable, limited bronzes/grays/black, energy-intensive, alloy-sensitive), and **two-step electrolytic** (durable, consistent, the architectural standard) — and say when each is chosen.
6. **Explain dimensional growth** — the coating grows roughly half in / half out; because architectural two-step coatings are *thick* (Class I), the growth is real and must be planned for on tight-fit features.
7. **Place two-step color in the family** — Type I (chromic), Type II (sulfuric, dyeable), Type III (hardcoat), and the specialty routes (BSAA, PAA, bright, integral color, **two-step electrolytic**).
8. **Explain why alloy/temper matters *less* here than in integral color** — because you deposit the color yourself in Step 2, two-step is **far more forgiving of alloy and lot variation**, which is the main reason it out-competes integral color on repeatability.
9. **Name the stations of the line in order** and explain *why* the order can't be rearranged (and why there are **two** main tanks).
10. **Recognize the main hazards** — Step-1 sulfuric acid, hot caustic etch, flammable hydrogen, high-current DC, near-boiling seal; **Step-2 AC current in an acidic metal-salt bath, and the toxicity of tin/nickel/cobalt salts (nickel and cobalt sensitizers/carcinogen concern by inhalation)** — and know the correct PPE, emergency response, and metal-salt waste handling.

★ **REMEMBER**

You are not expected to hit all of these on day one. Anodizing is a craft. But the *safety* objectives (#10) are not optional and not gradual — you must have those before you work the line at all.

• THE LINE AT A GLANCE

Don't memorize yet — feel the rhythm: **Inspect/Rack → Clean → Etch (satin) → Rinse → Desmut → Rinse → CLEAR-ANODIZE (Step 1, DC) → Rinse → AC COLOR (Step 2, metal salts) → Rinse → Seal → Rinse/Dry → Inspect.**



★ **REMEMBER**

Notice the pattern — **a rinse follows almost every working tank**, and there are now **two** rinses that matter most: the one *after* the clear anodize (keep the pores open and clean for coloring) and the one *after* coloring (before seal). Rinses aren't breaks; they're checkpoints that keep one chemistry from poisoning the next.

☀ **FUN FACT**

The aluminum oxide we grow in Step 1 is chemically the same family as **sapphire and ruby** (crystalline Al_2O_3). In Step 2 we tuck microscopic specks of **metal** down into that ceramic — so a two-step bronze is, quite literally, metal hidden inside a gemstone-family coating.

CHAPTER 1

WHAT IS ANODIZING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT ANODIZING IS

THE ONE-SENTENCE VERSION

Anodizing is using electricity to grow a hard, protective oxide layer out of the surface of an aluminum part — the part itself becomes part of the coating. That's the whole idea. Now let's unpack it, because it's genuinely different from plating and that difference trips up almost everyone at first.

FIRST, WHAT PLATING DOES (SO YOU CAN SEE THE DIFFERENCE)

In electroplating — zinc, nickel, chrome — you take your part, hook it up as the **cathode** (the *negative* electrode), and you *deposit a different metal* onto it from a bath. The coating is a *separate metal sitting on top of* your part.

Anodizing flips almost all of that:

- Your part is the **anode** — the **positive** electrode. (That's literally where the word "anodize" comes from.)
- We don't add a different metal. We take the **aluminum that's already there** and convert its surface into **aluminum oxide** by feeding electricity through an acid bath.
- The coating isn't sitting *on* the part — it **is** the part's surface, chemically grown and locked in. It can't peel like paint or flake like a bad plate, because it's integral to the metal.

> ★ **REMEMBER — THE SINGLE BIGGEST IDEA IN THIS BOOK**
> In plating, the part is the **cathode** and you **add** a metal. In anodizing, the part is the **anode** and you **grow oxide out of the part itself.** If you remember nothing else from Chapter 1, remember this. Everything else follows from it.

> ⚙ **TECHNICAL STUFF — A HEADS-UP ABOUT STEP 2**
> There is one important wrinkle on *this* line. In **Step 1** the part is the anode and we grow oxide (true anodizing). In **Step 2** — the coloring tank — we run **AC**, so the part is *alternately* anode and cathode many times a second, and on the **cathodic** half-cycles we actually **reduce metal ions and deposit metal** in the pore bottoms (a plating-like action). So this line is unusual: it does *both* — grow oxide first (DC anodize), then deposit metal (AC color). Hold that thought; Chapter 4 explains it in full.

HOW IT PHYSICALLY WORKS: THE KITCHEN-SINK PICTURE

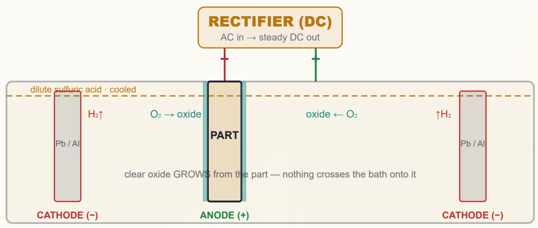
Picture a tank full of liquid called the **electrolyte** (or just "the bath"). For Step 1, that liquid is **dilute sulfuric acid** — the same as a Type II tank. You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction "DC" current anodizing needs):

- **The anode** — this is your **aluminum part**, connected to the **positive** terminal. *"Anode = the part, the thing growing oxide."*
- **The cathode** — bars or plates (commonly **lead** or **aluminum 6063**) hanging in the same tank, connected to the **negative** terminal. The cathode just completes the circuit and is where **hydrogen gas** bubbles off.

> ★ **REMEMBER**
> Step 1 of a two-step line is an **ordinary clear sulfuric (Type II) anodize** — nothing exotic. The clear, porous oxide it grows is simply the **home** the color metal will move into during Step 2. Get the clear coat right and the color follows.

When you turn on the rectifier (in Step 1), here's the dance that happens:

1. Current flows. At the surface of your aluminum part (the anode), water in the bath is split and **oxygen** is delivered right at the metal surface.
2. That oxygen immediately combines with the surface aluminum to form **aluminum oxide (Al_2O_3)** — a hard, ceramic layer.
3. The sulfuric acid is slightly aggressive, so as the oxide forms, the acid is *also* gently dissolving it — and this tug-of-war creates the famous **porous structure** (Chapter 4). Here those pores are the **home** for our color metal.
4. Over at the cathode, **hydrogen gas** bubbles off into the air (a safety item — flammable gas — we'll come back to it).



Step 1 looks like any Type II tank — the part is the **ANODE (+)** and oxide grows from it. The color comes later, in a separate tank.

⚙ TECHNICAL STUFF — THE TWO STEP-1 REACTIONS

At the part (anode), aluminum is oxidized: simplified, $2Al + 3H_2O \rightarrow Al_2O_3 + 6H^+ + 6e^-$. At the cathode, those electrons reduce hydrogen ions: $6H^+ + 6e^- \rightarrow 3H_2\uparrow$. So **oxygen builds the clear coating at your part; hydrogen gas leaves at the counter-electrode.** No aluminum travels through the bath onto something else — it stays put and turns to oxide in place.

⚙ TECHNICAL STUFF — WHY THE STEP-1 BATH MUST BE COOLED

The reaction is **exothermic** (it makes heat), and a lot of electrical energy turns to heat right at the part surface. If the bath warms, the acid dissolves the growing oxide too fast and you get a soft, powdery, "burned" coating — which then colors **unevenly** in Step 2. That's why a Step-1 tank has **refrigerated cooling and air agitation** to hold ~20 °C (68 °F). On this line, *a uniform, well-grown clear coat is the foundation of a uniform color.*

☀ FUN FACT

Anodizing is one of the few finishing processes where the coating is *grown from* the part rather than *added to* it — which is exactly why a two-step bronze can't "chip off" to bare aluminum the way painted bronze can. The oxide is continuous with the metal, and the color metal is locked deep inside it.

CHAPTER 2

WHAT IS TWO-STEP ELECTROLYTIC COLOR?

THE MODERN ARCHITECTURAL COLOR STANDARD — DURABLE, LIGHTFAST, CONSISTENT, AND FAMOUS FOR ITS BLACKS AND BRONZES

Two-step electrolytic color anodizing (also called **two-step color**, **electrolytic coloring**, or **electrocoloring**) is exactly what the name says — **two operations:**

1. **Step 1 — Clear anodize.** Grow an ordinary **clear sulfuric (Type II) anodize**: a hard, porous aluminum-oxide coating, typically run to **architectural Class I thickness (≥18 µm / 0.7 mil)**, with the part as the anode under DC. No color yet — the part comes out clear/natural.
2. **Step 2 — AC electrolytic coloring.** Move the part to a **second tank** containing dissolved **metal salts** — most commonly **tin (stannous sulfate)**, sometimes **nickel sulfate** or **cobalt sulfate** — in dilute sulfuric acid. Apply **AC (alternating current), typically ~10–20 V AC, for seconds to several minutes.** On the cathodic half-cycles, the metal ions are **reduced and deposited as fine particles deep in the bottom of the pores.** That buried metal **scatters and absorbs light**, producing **champagne** → **light bronze** → **bronze** → **dark bronze** → **black.**

Then the part is **rinsed and sealed**, exactly as on any anodize line, to close the pores and lock everything in.

> ★ **REMEMBER**
> The defining feature is in the name — **two steps, two tanks.** Step 1 makes the home (the porous clear oxide); Step 2 moves the color metal in (AC deposition at the pore base). **Color depth is set in Step 2 by how long and how hard you color** — *not* by how thick you grew the anodize. Hold onto that; it's the difference from integral color.

• WHY SHOPS (AND ARCHITECTS) CHOOSE IT

ADVANTAGE	WHAT IT MEANS ON THE FLOOR / ON THE BUILDING
Outstanding lightfastness & durability	Color is durable metal buried in the ceramic, not a dye — it won't fade in decades of UV. The reason it's specified for exterior architecture.
More energy-efficient than integral color	The color step is a short, **low-energy AC dip** instead of a thick, high-voltage, heavily-cooled "self-coloring" anodize.
Excellent consistency & repeatability	You *deposit* the color yourself, so it's far less dependent on the exact alloy/heat/lot — the central win over integral color.
Superb blacks and bronzes	Tin coloring gives a wide, clean tonal range and the **best deep blacks** in the anodize world.
Color can't peel or flake	The metal is locked deep in the pores, then sealed shut.
Meets architectural specs	Class I (≥18 µm) clear coat + electrolytic color satisfies **AAMA 611** and **ASTM B580** for long exterior service.

WHERE YOU'LL SEE IT

Building **curtain walls and facades**, **storefronts and entrances**, **window and door frames and systems**, **column covers, fascia, louvers, sunshades, railings, and architectural hardware** — anywhere a colored aluminum finish has to look right after decades of sun and weather. If you've stood in front of a **dark-bronze or black aluminum** building put up in the last few decades, you've very likely seen two-step electrolytic color.

> 💡 **TIP**
> When a print or architectural spec calls for "**dark bronze / statuary bronze / black anodize, AAMA 611, Class I**" with no mention of dye, it is very often asking for **two-step electrolytic color** (or, on older/legacy jobs, integral color) — *not* dyed Type II. Dyed bronze exists but is not durable enough for exterior architecture. **Check the spec and ask** — chemistry, cost, and durability all change.

> ☀ **FUN FACT**
> Electrolytic coloring grew up in the 1960s–70s (an early commercial route was the Italian **"Anolok"/Anodal** tin process) precisely *because* integral color was so power-hungry and so fussy about the alloy. Giving the operator a separate, controllable color tank turned "the metal makes the color" into "**we** make the color" — and the architecture industry never looked back.

CHAPTER 3 · READ TWICE

THREE WAYS TO COLOR ANODIZED ALUMINUM

THE ONE CHAPTER THAT MAKES SENSE OF WHERE THIS PROCESS SITS — DYE VS. INTEGRAL VS. TWO-STEP

Anodized aluminum can be colored **three fundamentally different ways**, and they are not interchangeable.

ROUTE 1 — DYEING (TYPE II + ORGANIC DYE IN THE PORES)

Anodize a normal clear coating, then soak the **open pores** full of **organic (or inorganic) dye**, then seal.

- **Pros:** Cheap, fast, **any color** including brilliant reds, blues, golds. Low energy.
- **Cons: Least lightfast** — many organic dyes fade in UV/outdoors. Color sits in the pores, not locked as durable metal. Best for **indoor / decorative / consumer** work.

ROUTE 2 — INTEGRAL COLOR (COLOR FROM THE ANODIZING ITSELF)

Anodize in a **sulfuric + organic acid** bath at **higher voltage**; the **alloy constituents + the modified oxide make the color as the coating grows.** No dye, no second tank.

- **Pros: Extremely lightfast and abrasion-resistant** — color is in the ceramic.
- **Cons: Limited palette** (champagne, bronzes, gray, black). **Energy-intensive** (high voltage, long cycle, big cooling load). **Very alloy-sensitive** — the metal *is* part of the colorant, so **lot-to-lot color matching is hard.**

ROUTE 3 — TWO-STEP ELECTROLYTIC COLOR (THIS MANUAL)

Anodize a normal **clear** coating (Step 1), then move to a **second electrolytic tank** of **metal salts** (tin/nickel/cobalt) and apply **AC** to **deposit fine metal deep in the bottom of the pores** (Step 2). Then seal.

- **Pros: Durable and lightfast** like integral color (the color is metal, not dye) — **but far more consistent, lower energy, and much less alloy-sensitive.** Tin gives champagne→bronze→**black** with excellent depth.
- **Cons:** Two process steps and a second tank to run/maintain; palette is mostly **bronzes/black/gray** (some blues/burgundies possible with special processes), so it isn't the route for a bright primary color.
- **This is the modern architectural standard** and has **largely displaced integral color** for new work — it's cheaper to run and more repeatable.

> ★ **REMEMBER — THE THREE ROUTES IN ONE BREATH**
> **Dye = color *in the pores* (cheap, any color, fades). Integral = color *in the oxide itself* (durable, bronzes/black, energy-hungry, alloy-fussy). Two-step = metal *deposited in the pore bottoms* with AC (durable, consistent, energy-efficient — the modern architectural default, with the best blacks).**

	DYE (TYPE II)	INTEGRAL COLOR	TWO-STEP ELECTROLYTIC
Where the color lives	Organic dye **in the pores**	**In the oxide itself** (alloy + structure)	**Metal particles deep in the pore bottoms**
How color is made	Soaked-in dye	Grown during anodize (thickness/alloy)	**AC deposition in a metal-salt tank**
Lightfastness	Low–medium (UV fades)	**Excellent**	**Excellent**
Palette	**Any color**	Champagne, bronzes, gray, black	Champagne → bronze → **black** (some others)
Energy / cost	Low	**High** (high V, long cycle, cooling)	**Moderate** (short low-energy AC step)
Color consistency	Good (dye-controlled)	**Hard** (alloy-driven)	**Very good** (you deposit it)
Alloy sensitivity	Low–moderate	**High**	**Low**
Typical use	Indoor / consumer / decorative	Exterior architecture (legacy & spec)	**Exterior architecture (modern std)**

TECHNICAL STUFF — WHY TWO-STEP IS SO MUCH LESS ALLOY-SENSITIVE

In integral color, the alloy's own intermetallic particles *are* part of the colorant, so a different heat or extrusion changes the color at identical settings. In two-step, the color comes from **metal you deposit yourself** in a separate, controllable tank — so the base alloy matters far less, and **lot-to-lot color matching is dramatically easier.** That single fact is the core reason the industry drifted from integral toward two-step: **you control the color directly instead of relying on the metal to make it for you.**

FUN FACT

All three routes can produce a "bronze" that looks similar on day one. Put them in the desert sun for ten years and they sort themselves out: the **dyed** one fades first, while the **integral** and **two-step** bronzes hold — because in those two the color is durable metal/ceramic, not an organic dye.

TIP — PICK THE ROUTE TO THE REQUIREMENT

Bright color indoors → **dyed Type II.** Durable bronze/black outdoors → **integral or two-step.** When cost, energy, **lot-to-lot consistency, and deep blacks** matter — which is most modern architectural work — **two-step wins.** Knowing which route fits the job is half of running this line well.

CHAPTER 4 · READ TWICE

COLOR IN THE PORE BOTTOMS

THE STRANGEST, MOST IMPORTANT IDEA ON THIS LINE — YOU PLANT METAL IN THE PORES WITH ALTERNATING CURRENT

This is the chapter that makes two-step color two-step color. Slow down here.

SAME POROUS OXIDE — BUT NOW IT'S A HOME FOR METAL

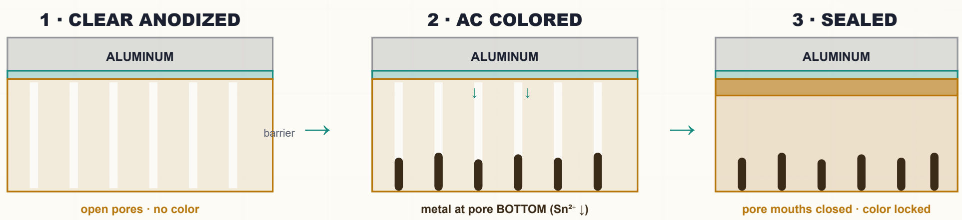
Like all sulfuric anodizing, Step 1 grows a **porous oxide**: billions of tiny columns, each with a microscopic **pore** down the center, sitting on a thin, dense **barrier layer** against the metal. On a *dyed* line we'd fill those pores with dye. Here we do something different — **we plant metal at the very *bottom* of the pores, right above the barrier layer.**

STEP 2: AC DRIVES METAL INTO THE PORE BOTTOMS

Move the freshly anodized (still **clear, unsealed, pores-open**) part into the **coloring tank** — dilute sulfuric acid carrying dissolved **metal salt** (tin/nickel/cobalt). Now apply **AC (alternating current)**, with the part on one leg and inert counter-electrodes (graphite/carbon or stainless steel) on the other. Here's the dance:

- On the **cathodic half-cycle** (part momentarily negative), metal ions (e.g., **Sn^{2+}**) are pulled down the pores and **reduced to solid metal**, depositing as fine particles **at the pore base.**
- On the **anodic half-cycle** (part momentarily positive), very little happens to that buried metal — the barrier layer's one-way ("valve/rectifier") behavior largely blocks redissolving it.
- Cycle after cycle (50/60 times a second), the metal **accumulates into tiny columns at the bottom of each pore.**

That buried metal **absorbs and scatters light** coming into the coating. **A little metal = champagne/light bronze; more metal = medium and dark bronze; a lot of metal = black.**

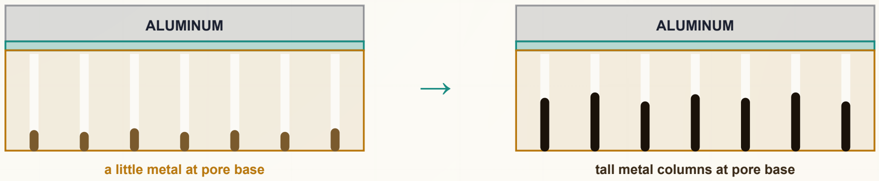


Color metal sits at the pore **BOTTOM**, not the top — which is why it's so durable and so well protected once sealed.

COLOR DEPTH = COLORING TIME × AC VOLTAGE — *NOT* THICKNESS

Here is the operator's golden rule of this line: **the more metal you deposit, the darker the color — and you deposit more metal by coloring longer and/or at higher AC voltage.** A short dip = champagne; a longer dip at higher volts = black. **The clear-anodize thickness is held essentially constant** (architectural Class I), and **the color is dialed in entirely at Step 2.**

1 · SHORT TIME / LOWER V = LIGHT BRONZE **2 · LONG TIME / HIGHER V = DARK / BLACK**



Same clear coat — color deepens with more coloring time/voltage (more deposited metal). Contrast this with integral color, where darker means a **thicker** coating.

APPROX. COLORING TIME/VOLTAGE (ILLUSTRATIVE)	TYPICAL TIN (SN) COLOR
`~20–60 s, low V`	Champagne / very light bronze
`~1–3 min`	Light to medium bronze
`~3–6 min`	Medium to dark bronze
`~6–10 min`	Dark bronze / charcoal
`~10–15 min, higher V`	**Black**

Representative only. Exact time/voltage-to-color depends on the specific coloring chemistry, metal concentration, temperature, and your clear-coat — **always color a sample and run to your shop's color standard and ΔE limit.**

★ REMEMBER — THE HEART OF TWO-STEP COLOR

Color is deposited metal. Darker = more metal = more coloring time and/or higher AC voltage. The clear-anodize thickness is held to spec and is *not* the color knob — the **color tank** is. This is the key contrast with integral color, where darker meant a thicker coating on a tightly-controlled alloy.

⚠ HEADS-UP — CATCH COLOR BEFORE YOU SEAL

Once a part is **sealed**, you can't add more color (the pores are shut). The whole point of inspecting color *before* sealing is that a too-light part can often go **back into the color tank** for more time. A too-dark or sealed-wrong part must be **stripped and re-anodized** — expensive and dimension-eating. **Color to standard, verify, *then* seal.**

⚙ TECHNICAL STUFF — WHY AC AND NOT DC

The dense **barrier layer** at the pore bottom acts like a **rectifier (a one-way valve).** Under plain DC, it mostly just keeps re-anodizing and won't let you build metal cleanly. **AC** exploits the asymmetry: the cathodic half-cycle pushes metal *in* and reduces it at the pore base, while the anodic half-cycle is largely blocked from removing it — so metal **accumulates** deep in the pore. (Some modern processes use shaped/asymmetric or pulsed AC for finer control — read your process TDS.)

☀ FUN FACT

Because the color metal lives at the **bottom** of the pores — protected by the whole depth of the oxide above it *and* the seal — two-step colors are extraordinarily abrasion- and weather-resistant. You can scuff the surface and barely touch the color, because the color isn't on top; it's at the very floor of the pore.

CHAPTER 5

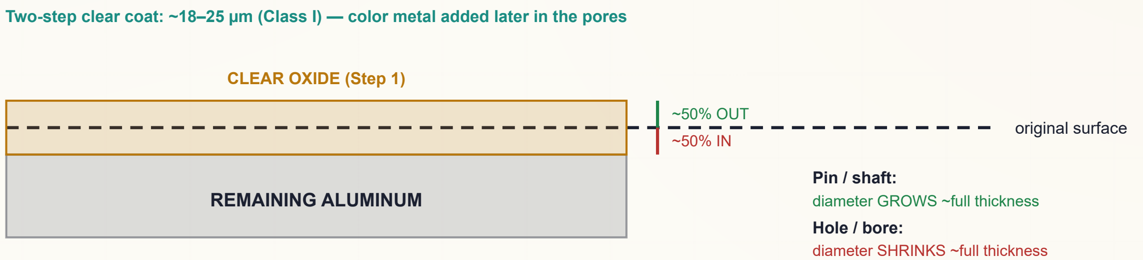
DIMENSIONAL GROWTH

THE COATING MAKES THE PART BIGGER — AND ON A THICK ARCHITECTURAL FILM, YOU MUST PLAN FOR IT

Because we're converting aluminum into aluminum oxide, and oxide takes up **more room** than the metal it came from, the **clear coating** (Step 1) grows in **two directions at once**:

- About **half of the coating thickness grows *into*** the original surface (consuming aluminum).
- About **half grows *out*** from the original surface (the part gets bigger).

So for a Class I architectural coating of, say, **20 µm (0.0008″)**, the surface moves outward by roughly **10 µm (0.0004″) per face.** On a shaft/pin the **diameter grows by about the full coating thickness**; a hole's **diameter shrinks by about the full coating thickness.** The Step-2 color metal adds **negligible** dimension — essentially all the growth is from the Step-1 oxide.



★ REMEMBER — THE 50% RULE

Anodize grows **roughly half in, half out.** Each *surface* moves out ~half the coating thickness; a *diameter* changes by ~the full coating thickness. Because two-step color is run **thick (Class I, ≥18 µm)** for durability, this growth is **not negligible** — tight-fit features must account for it *before* anodizing. **The coloring step adds essentially nothing dimensionally — it's the Step-1 oxide that grows.**

💡 TIP

Architectural extrusions usually have generous tolerances, but **interfaces matter:** snap-fits, screw bosses, weep slots, glazing pockets, and thermal-break channels can be thrown off by a thick coating. Rule of thumb: **~half the spec'd clear-coat thickness per surface** — confirm against the drawing and your shop standard.

⚠ HEADS-UP — MASKING & GROUNDING

Surfaces that must *not* grow or must stay conductive (grounding lugs, screw threads, electrical bonds) are **masked** or machined to allow for growth. Anodize is **electrically insulating** and a Class I film is *thick* — a ground path left unmasked will definitely not conduct afterward. **Masking also matters for color uniformity:** a poorly masked contact or shadow can color unevenly in Step 2.

⚙ TECHNICAL STUFF — PILLING–BEDWORTH

Aluminum oxide occupies more volume than the aluminum it formed from (Pilling–Bedworth ratio > 1). That expansion is *why* the coating grows outward at all — and it's the same expansion that lets hot-water sealing pinch the pores shut later (with the color metal safely sealed inside).

☀ FUN FACT

The growth across a wide architectural face is still just microns — invisible to the eye — yet enough to bind a precision fit. The coating you can barely measure can still seize a tight assembly.

CHAPTER 6

THE ANODIZE FAMILY

SAME IDEA, MANY FLAVORS — KNOW WHICH ONE YOUR PRINT IS ASKING FOR

All anodize “types” grow an integral aluminum-oxide coating with the part as the anode. They differ in **acid, thickness, and purpose** — and the *coloring routes* sit on top of that.

	TYPE I — CHROMIC	TYPE II — SULFURIC	TYPE III — HARDCOAT	TWO-STEP COLOR *(THIS MANUAL)*
Electrolyte (Step 1)	Chromic acid	Sulfuric acid	Sulfuric (colder/stronger)	**Sulfuric acid (clear), then a separate metal-salt color tank**
Typical thickness	~0.5–7.5 µm	~5–25 µm	~25–100+ µm	**~18–25 µm (Class I) clear coat**
Color source	Gray, limited	Dye (in pores)	Darkens naturally; dyeable	**Metal deposited in pore bottoms (AC)**
Lightfastness of color	n/a	Dye-dependent	Good	**Excellent**
Main use	Aerospace (fatigue)	Decorative + protective workhorse	Max wear/hardness	**Exterior architectural color**

★ **REMEMBER**

Type I = chromic, thin, aerospace. Type II = sulfuric, the dyeable workhorse. Type III = hardcoat, thick, max wear. Two-step color is **built on a Type II clear anodize** plus a second **AC metal-coloring** step — it’s a *coloring route*, not a new “type.” On a print you’ll often see it as a clear-coat *type/class* (e.g., Type II, Class I) **plus** a color/spec callout (e.g., AAMA 611 dark bronze).

💡 **TIP**

The other specialty anodizes in this training series — **BSAA** (boric-sulfuric, aerospace chromic-replacement), **PAA** (phosphoric, for adhesive bonding), **bright anodize** (polished first), and the two color routes **integral color** and **two-step electrolytic** — all stand on the same Type II fundamentals. Each has its own manual; they all share the part-is-the-anode / porous-oxide / seal core.

⚙ **TECHNICAL STUFF — WHERE TWO-STEP SITS ENERGETICALLY**

Integral color is run like a “self-coloring hardcoat” — high voltage, long cycle, heavy cooling — so it’s **energy-hungry.** Two-step keeps **Step 1 as an ordinary Type II anodize** and adds a **short, low-voltage AC color dip.** Net result: a comparable durable color for **less total energy** — one of its headline advantages.

☀ **FUN FACT**

“Two-step” really is just “Type II, then a color tank.” Everything you already know about an ordinary sulfuric anodize line still applies to Step 1 — the only genuinely new tank to learn is the AC color tank at Station 09.

CHAPTER 7

ALUMINUM ALLOY & TEMPER

LESS SENSITIVE HERE, AND THAT'S THE POINT — YOU DEPOSIT THE COLOR YOURSELF

Anodizing only works on **aluminum** and a few other "valve metals" (titanium, magnesium, tantalum, niobium — each with its own process). On this line it's aluminum — but "aluminum" is dozens of **alloys**, and the alloying elements change how a part anodizes.

THE FRIENDLY ALLOYS (AND THE TROUBLEMAKERS)

- **6xxx (Mg-Si, esp. 6063 "architectural")** — anodizes clean and bright; the everyday favorite for architectural extrusions and the backbone of two-step color work.
- **5xxx (Mg)** — anodizes very well (common for sheet/panel).
- **Copper (2xxx)** — dull/yellowish, harder to anodize well (rare in architecture).
- **Silicon (cast, high-Si)** — gray/mottled clear coat; colors unevenly. Architectural work avoids it.

ALLOY FAMILY	MAIN ELEMENT	CLEAR-ANODIZE BEHAVIOR
1xxx (pure Al)	none (99%+ Al)	Excellent, very clear
5xxx	Magnesium	Very good, clear
6xxx (6061/6063)	Mg + Si	**Excellent — the architectural workhorse**
7xxx	Zinc	Good, can vary
2xxx	Copper	Dull/yellow, harder
Cast (high-Si)	Silicon	Gray/mottled, colors poorly

THE BIG SELLING POINT: TWO-STEP IS MORE FORGIVING

Here is the crucial contrast with integral color. In **integral color**, the alloy *is* the colorant, so a different heat or extrusion changes the bronze at identical settings — lot-to-lot matching is the central headache. In **two-step**, you **deposit the color metal yourself** in Step 2, so the base alloy's exact chemistry matters **much less.** Two extrusions from slightly different heats of 6063 will color **far more alike** in a two-step tank than in an integral tank.

★ **REMEMBER**

Two-step color is the *least* alloy-sensitive of the durable coloring routes. That forgiveness — plus lower energy — is exactly why architects and anodizers moved to it. **You still want a clean architectural alloy (6063/6061/5xxx) for a uniform clear coat**, but you no longer live or die by matching the metal heat the way integral color does.

💡 **TIP**

Even though two-step is forgiving, **still rack like alloy with like alloy** and **run a sample of the actual metal** for tight color jobs. "Less sensitive" is not "not sensitive" — surface finish, etch, and clear-coat uniformity still drive the final ΔE.

☀ **FUN FACT**

Pure and high-purity 6063 anodize so cleanly that the architectural industry built whole product lines (window frames, storefront systems) around them — and two-step coloring is what lets those same extrusions come out a consistent dark bronze or black across an entire building.

CHAPTER 8

HOW THE LINE WORKS

WALKING THE LINE — EVERY TANK HAS A JOB, AND NOW THERE ARE TWO MAIN EVENTS

A two-step color line is a sequence of tanks. Here's the journey and *why each step is where it is.*

1. **Inspect / Rack** — Check the part, confirm alloy and spec (Class I, target color/ΔE), and clamp it tight. The part **is the anode**, so contact is everything.
2. **Clean / Degrease** — Remove oils and soil; oxide won't grow evenly through grease.
3. **Etch (caustic, architectural satin)** — Remove the natural oxide skin and a little aluminum to give the uniform **satin/matte** architectural finish; sets the look the color sits on.
4. **Rinse** — Wash caustic off before the acid steps.
5. **Desmut / Deox** — Remove the dark smut the etch leaves; a uniform, bright surface is essential for **even color.**
6. **Rinse** — Clean handoff to the anodize tank.
7. **CLEAR ANODIZE — Step 1** — Part = anode, dilute sulfuric, cooled, DC on. Grow the **clear** Class I porous oxide.
8. **Rinse** — Gently rinse acid out of the **open pores** with **cool DI** — do *not* pre-seal, and do *not* contaminate, or the color goes uneven.
9. **AC COLOR — Step 2** — Metal-salt tank, **AC** on. Deposit color metal in the pore bottoms to the target shade/ΔE.
10. **Rinse** — Remove color-bath drag-out before seal.
11. **Seal** — Close the pores to lock in color metal and corrosion resistance.
12. **Rinse / Dry** — Final clean-up and gentle dry.
13. **Inspect** — Color/ΔE, thickness, seal quality, appearance.

★ **REMEMBER — WHY THE ORDER CAN'T MOVE**

You can't color before you anodize (no pores yet). You **must keep the pores open** between anodize and color (no pre-sealing). You can't seal before you color (sealed pores won't take the metal). You can't anodize over grease or smut (uneven oxide → uneven color). **The line is the recipe, and the two main events — clear anodize, then AC color — happen in that order for a reason.**

⚠ **HEADS-UP — THE LINE IS A TOUR OF HAZARDS, NOW WITH A COLOR TANK**

As you walk the line you pass **hot caustic** (etch), **strong acids** (desmut, anodize, color bath), **flammable hydrogen** (Step-1 cathode), **high-current DC** (Step-1 rectifier), **AC current in an acidic metal-salt solution** (Step-2 color tank), **near-boiling water** (seal), and **toxic metal salts** (tin/nickel/cobalt). Each station chapter calls out its own. Respect each tank for what *it* is.

💡 **TIP**

When you're learning, walk the empty line with your trainer and name each tank's *job* and *main hazard* out loud — and especially be able to say which two tanks are the "main events" (clear anodize, AC color) and what each one does.

CHAPTER 9

PPE — YOUR DAILY ARMOR

WHAT YOU WEAR, AND WHY EACH PIECE IS THERE

A two-step line works with strong sulfuric acid, hot caustic etch, near-boiling seal tanks, high-current DC, **AC current in an acidic metal-salt color bath**, chemical mist, and **tin/nickel/cobalt salts.** None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

HAZARD ON THE LINE	PPE THAT PROTECTS YOU
Sulfuric acid (anodize, desmut, color bath) — burns, splash	Sealed chemical **splash goggles** + **face shield**, chemical-resistant **gloves**, **apron/suit**, chemical-resistant **boots**
Hot caustic etch — deep burns	Same as above; caustic burns are deep and slow to feel at first
Near-boiling seal tank (~96–100 °C) — scald/steam	Face shield, heat-resistant gloves, long sleeves; mind the steam
Acid mist / fumes; metal-salt mist (nickel/cobalt)	**Local exhaust at tanks**; respiratory protection per your program — **avoid breathing color-bath mist**
DC (Step 1) and AC (Step 2) electrical	Dry hands, dry gloves, no jewelry, follow LOTO; never reach into an energized tank
Tin/nickel/cobalt salts — sensitizers; Ni/Co inhalation carcinogen concern	Gloves, goggles; no skin contact; **no hand-to-mouth**; wash thoroughly
Slips on wet floors	Non-slip chemical-resistant boots

💡 **TIP**

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. Rinse your gloves *before* you take them off — especially after the color tank, where they carry metal salts.

★ **REMEMBER**

On this line your **eyes, skin, and lungs** are all exposed. Strong acid *and* hot caustic *and* near-boiling water *and* a metal-salt color bath all live here. **Sealed splash goggles and a face shield are not optional** at the working tanks, and **local exhaust is your first defense against nickel/cobalt mist.**

⚠ **HEADS-UP — NO EATING/DRINKING/VAPING ON THE LINE**

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting chemicals — **especially nickel/cobalt color salts and nickel-bearing seal salts.** Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

⚠ **HEADS-UP — GOGGLES, NOT JUST GLASSES**

Safety glasses stop flying chips; they do **not** stop an acid or caustic splash from the side. At every chemical tank — including the AC color tank — sealed splash goggles (and a face shield when charging, dumping, or handling parts) are required.

CHAPTER 10

EMERGENCY & FIRST-RESPONSE

KNOW THESE COLD BEFORE YOU TOUCH A TANK

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read.

SITUATION	WHAT TO DO — IMMEDIATELY
Acid or caustic on skin	Flush with water **at least 15 minutes.** Remove contaminated clothing while flushing. Get medical attention — caustic burns can be deep even when they don't hurt at first.
Acid or caustic in eyes	Go straight to the **eyewash**, hold eyes open, flush **at least 15 minutes**, get medical help. Don't stop early.
Large splash / drench	**Safety shower** — full 15 minutes, clothing off.
Hot seal-tank scald	Cool with water; don't pop blisters; get medical attention for anything beyond minor.
Color-bath (metal-salt) splash	Flush skin/eyes **15 minutes**; the solution is **acidic and metal-bearing** — get medical help for eye contact; wash thoroughly to limit nickel/cobalt sensitization.
Acid/metal-salt spill	Don't play hero — alert others, contain only if trained, use the spill kit, follow your shop's plan. Never casually mix acids and caustics. **Metal-salt spills go to treatment, not the drain.**
Electrical / arc at rectifier, bus, or AC color supply	Don't touch — de-energize via the controls / **LOTO.** Never reach into a tank with current on.
Hydrogen gas (Step-1 cathode)	Keep ignition sources (flames, sparks, grinding) away; ensure ventilation is running.
Fluoride exposure (fluoride desmut / cold seal)	Fluoride burns are deep, delayed, and systemically toxic — flush, then apply the **specific first-aid** in the SDS (often **calcium gluconate gel**) and get medical help immediately. Ordinary flushing alone may not be enough.

⚠ HEADS-UP — THE GOLDEN RULE FOR ACID MAKE-UP

Always add ACID to WATER, never water to acid. ("Do as you *oughta*: add acid to water.") Adding water to concentrated sulfuric can flash-boil and erupt. Bath make-up/additions are usually authorized-personnel work — but every operator must know this rule.

⚠ HEADS-UP — LOCKOUT/TAGOUT (LOTO)

LOTO is mandatory for *all* electrical work — the **Step-1 DC rectifier** *and* the **Step-2 AC coloring supply.** Physically lock the power off and tag it. "I'll just be a second" is how people get electrocuted — and the color tank's AC is just as dangerous as the anodize tank's DC.

★ REMEMBER

Know where the **nearest eyewash, safety shower, and spill kit** are at *every* station before your shift. Eyewash and shower stations must always be **unblocked.**

CHAPTER 11

THE SAFETY PHILOSOPHY OF A TWO-STEP COLOR LINE

THE MINDSET THAT TIES EVERY STATION TOGETHER

1. TWO CORROSIVES, MIRROR-IMAGE.

The **etch is caustic** (high pH); the **desmut, anodize, and color bath are acid** (low pH). Both burn. Treat high pH and low pH with equal respect — and never let them mix outside controlled neutralization.

2. THE PART IS LIVE — TWICE.

In Step 1 the part is the anode under **DC**; in Step 2 it carries **AC** in a conductive acid solution. Wet hands + current + acid is a bad combination at *both* main tanks.

3. NEW ON THIS LINE: TOXIC COLOR METALS.

The color bath holds **tin, nickel, or cobalt salts.** Tin is comparatively mild; **nickel and cobalt are skin sensitizers and carcinogen concerns by inhalation.** Control the **mist** (exhaust), keep it off your **skin**, and never let it reach your **mouth.**

4. HOT IS A HAZARD TOO.

The seal tank runs **near boiling.** Steam scalds.

5. HYDROGEN IS FLAMMABLE.

The Step-1 cathode bubbles off hydrogen. Ventilation on, ignition sources away. **Engineering controls first, PPE second** — tank exhaust, cooling, lids, splash guards do the heavy lifting; PPE is your last layer. And **when in doubt, stop and ask:** the unsafe operator isn't the one who asks questions — it's the one who guesses.

• THE HAZARD FAMILIES TO INTERNALIZE

1. **Corrosives (acid AND caustic).** Sulfuric (anodize/desmut/color) and caustic (etch). *Defense:* goggles, face shield, gloves, apron; 15-minute flush; acid-to-water.
2. **Toxic metals (tin/nickel/cobalt).** *Defense:* local exhaust, gloves, no hand-to-mouth, segregate/treat waste.
3. **Hot / thermal.** Near-boiling seal, hot etch. *Defense:* face shield, heat gloves, steam awareness.
4. **Electrical & gas.** DC (Step 1), **AC (Step 2)**; flammable hydrogen. *Defense:* LOTO on both supplies, dry hands, no ignition sources, ventilation.

★ **REMEMBER**

Corrosives, toxic metals, thermal, electrical/gas. If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself. **The color tank adds two families at once — metal toxicity *and* AC electrical — so treat it with full respect, not as "just a quick dip."**

★ **REMEMBER**

You now have the whole foundation: what anodizing is (and how it's *not* plating), the part-is-the-anode/grow-oxide idea, the two-step concept, color-metal-in-the-pore-bottoms via AC, the three coloring routes, dimensional growth, the family, alloys, how the line works, what to wear, and the safety mindset. Every station chapter builds on this. Turn the page knowing the *why*, and the *how* will make a lot more sense.

PART TWO — STATION BY STATION
STATION 01 OF 13

INSPECTION & RACKING

"IN ANODIZING, THE PART IS THE WIRE. A LOOSE PART IS A DEAD PART — AND ON A COLOR LINE, A STREAKY ONE."

PURPOSE & THE "WHY"

Inspect the incoming aluminum, confirm the **alloy and spec** (Class I clear-coat thickness, target **color/ΔE**, AAMA 611 or project spec), and **rack it** so it makes firm, current-carrying contact and hangs correctly. Because the part **is the anode** in Step 1 — and carries **AC** in Step 2 — every bit of current flows *through the rack contact into the part.* A loose contact means **no coating, a thin coating, or a burned spot** — and in Step 2 it means **uneven or no color** at that area. Racking also decides *where* the un-anodized contact mark lands, how gas escapes, and how solution drains.

★ **REMEMBER — THE CONTACT MARK IS UNAVOIDABLE**

Wherever the rack grips, **no oxide grows** (that spot must carry current). So place the contact where the mark won't show — a hidden face, an edge, a screw boss — and make it **tight.** On a color line, a poor contact shows up twice: a thin clear coat *and* a color miss.

ITEM	SPEC / PRACTICE	WHAT TO WATCH
Incoming inspection	Per traveler / print	Confirm **alloy**, Class I thickness, **target color/ΔE**, spec (AAMA 611)
Rack material	Titanium or aluminum	Ti racks don't anodize away; Al racks anodize and need stripping
Contact pressure	Firm, spring-loaded / bolted	Loose = burning, no coat, **color miss**, dropped parts
Contact location	Hidden/non-critical area	The contact point won't anodize or color (bare mark)
Masking	Plugs/tape/lacquer on no-anodize areas	Threads, grounds, bearing fits
Orientation	Allow gas escape & drainage	Avoid trapped air (un-anodized/un-colored voids) and color-bath pockets

⚠ **HEADS-UP**

Racks and parts have **sharp edges and pinch points**, handled near chemical tanks. A part that drops from a weak rack becomes a **splash hazard** in an acid, caustic, or **metal-salt** tank — not just scrap.

💡 **TIP — TITANIUM RACKS ARE WORTH IT**

Titanium racks don't anodize (they passivate), so they hold good contact load after load. Aluminum racks anodize along with the part and must be **stripped/re-pointed** regularly. Consistent contact = consistent thickness = consistent color.

STEP-BY-STEP

1. Read the traveler (alloy, Class I, target color/ΔE, masked areas). 2. Inspect for scratches/gouges — anodize *reveals* flaws and **color exaggerates them.** 3. Plan a hidden contact that carries full current. 4. Mask no-anodize areas. 5. Rack firmly, correct orientation. 6. Verify no parts touch (shadowing → color streaks). 7. Handle by the rack from here on.

SIGN-OFF — STATION 01

Teach-Back: why contact matters more in anodizing than plating (part = anode, and carries AC in Step 2); why the contact point won't anodize or color; why titanium racks are preferred; why orientation matters for gas/drainage; two no-anodize areas that get masked and why.

"I verified alloy/spec and target color, chose and made a firm contact in a non-critical area, masked all no-anodize features, racked for gas escape and drainage with no parts touching, and handled parts by the rack."

Trainee: ____________ Trainer: ____________ Date: ________

STATION 02 OF 13

CLEAN / DEGREASE

“ANODIZE OVER OIL AND YOU’VE GROWN A DEFECT, NOT A COATING — AND IT’LL COLOR BLOTCHY.”

PURPOSE & THE “WHY”

Get the part **chemically clean** — no oil, grease, fingerprints, or soil — so the oxide grows evenly and **colors uniformly.** Cleaning is usually a **mild alkaline (non-etching) soak cleaner** — formulated *not* to attack the aluminum (we save controlled metal removal for the etch). The free, foolproof check is the **water-break test.**

TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; use it on every part.

```
FAIL - water beads up          PASS - water sheets evenly
+--------------+               +--------------+
|  o   o   o   |  X            |##############|  OK
|    o    o    |               |##############|
+--------------+               +--------------+
Oil remains -> RE-CLEAN        Surface clean -> proceed
```

PARAMETER	RANGE	WHAT TO WATCH
Type	Mild **non-etching alkaline** cleaner	Must not attack aluminum
Temperature	~50–70 °C (120–160 °F)	Per cleaner TDS
Time	~3–10 min	Heavy oil longer
Concentration	Per TDS	Titrate regularly
Water-break test	Pass = continuous sheet	The only reliable field check

⚠ HEADS-UP — HOT ALKALINE SOLUTION

Even a “mild” cleaner is hot and alkaline: splashes irritate/burn skin and eyes; steam irritates airways. Splash goggles, gloves, apron. Skin → flush 15 min; eyes → eyewash 15 min and get help.

STEP-BY-STEP

1. Check tank temp/concentration in range. 2. Immerse fully; start timer; confirm agitation. 3. Withdraw slowly, draining back. 4. Run the **water-break test** — sheet → proceed; beading → re-clean. 5. Move promptly to etch.

SIGN-OFF — STATION 02

Teach-Back: what the cleaner removes vs. what it must *not* do; perform/read a water-break test; the caustic/irritant hazard and PPE; why clean surfaces color uniformly.

“The part passed the water-break test, the cleaner was in range and at temperature, and I wore required PPE.”

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 03 OF 13

ETCH (ARCHITECTURAL SATIN)

"WHERE THE PART GETS ITS ARCHITECTURAL SATIN LOOK — AND WHERE YOU'LL RESPECT CAUSTIC FOREVER."

PURPOSE & THE "WHY"

Remove the natural oxide skin and a thin layer of aluminum in a **hot caustic (sodium hydroxide) etch**, giving the uniform, matte **satin/matte architectural finish** most building aluminum has — and a fresh surface for anodizing. The caustic **dissolves a controlled amount of aluminum**, evening out die lines and machining marks and producing the soft, frosty appearance. Etch too little → shiny, blotchy; etch too much → lost dimension and detail. **The texture the etch leaves is the texture the color sits on**, so etch uniformity drives color uniformity.

★ **REMEMBER — ETCH SETS THE TEXTURE**

The **etch decides satin vs. shinier.** Light etch = more luster (and visible defects); heavy etch = more matte (and more metal lost). Most architectural two-step work is a **medium-to-matte satin.** Most pre-anodize **dimensional loss** happens here.

PARAMETER	RANGE	WHAT IT DOES / WATCH
Chemistry	**Sodium hydroxide**, often with additives	Dissolves oxide + aluminum
Concentration	~40–60 g/L NaOH	Higher = faster/more aggressive
Temperature	~50–65 °C (120–150 °F)	Hot — burns and speeds etch
Time	~1–5 min	Longer = more matte + more metal loss
Result	Uniform satin + **smut**	Smut is normal — removed at desmut

⚠ **HEADS-UP — HOT CAUSTIC IS ONE OF THE WORST BURNS ON THE LINE**

Caustic burns are **deep, get worse with time, and may not hurt much at first.** Full PPE: sealed goggles, face shield, elbow-length chemical gloves, apron, boots. Skin → flush **15+ minutes**; eyes → eyewash 15 min and get help. Never assume "it's just a little."

⚠ **HEADS-UP — CAUSTIC + ALUMINUM MAKES HYDROGEN**

Caustic attacking aluminum releases **hydrogen gas** (flammable). Keep ignition sources away and ventilation running.

STEP-BY-STEP

1. Confirm concentration/temp in range. 2. Immerse; start the timer for the **specified etch time** (controls finish, metal loss, and color uniformity). 3. Agitate per procedure. 4. Withdraw and drain; the part will look **dark/smutty** (normal). 5. Move **directly to rinse** — don't let caustic dry on the part.

SIGN-OFF — STATION 03

Teach-Back: what the etch removes and the finish it creates; the link between etch time and appearance/dimensional loss/color uniformity; why the part comes out dark/smutty and what fixes it; the caustic-burn hazard and first response.

"I etched for the specified time at correct concentration/temperature, understood the appearance, dimensional, and color effects, moved straight to rinse, and wore full PPE for hot caustic."

Trainee: ____________________ Trainer: ____________________ Date: _____________

STATION 04 OF 13

RINSE (THE FIREWALL)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

PURPOSE & THE “WHY”

Wash the **caustic film** off before the part reaches the acidic desmut/anodize chemistry. Carry caustic drag-out forward and it **neutralizes and contaminates the acid baths**, wastes chemistry, and causes staining. Cleanliness is tracked by **conductivity** (µS/cm) — *lower = cleaner.*

TIP — COUNTER-FLOW RINSING

The professional approach is a **counter-current** cascade — parts move forward, water flows backward, so the cleanest water touches the part last. Saves enormous water on a line with this many rinses.

PARAMETER	RANGE	NOTES
Temperature	`Ambient`	No heating needed
Time	`30–60 s with agitation`	Longer for cavities
Water source	`Municipal or DI`	DI preferred near anodize/color/seal
Conductivity	`< 500 µS/cm (target < 300)`	Monitor at shift start
Flow	`Continuous overflow / counter-current`	Prevents stagnation

⚠ HEADS-UP

This rinse carries dragged-in caustic and starts mildly alkaline. Goggles, gloves, non-slip footwear. **Wet floors and slips are the #1 everyday injury at rinses.**

DO

- Check conductivity at shift start; tell your lead if near/above limit.
- Transfer promptly from the etch — don’t let caustic dry.
- Immerse fully with agitation 30–60 s (longer for cavities).
- Lift and drain over the rinse tank before moving on.

TOP MISTAKES

- Treating the rinse as a quick dunk.
- Ignoring conductivity.
- Letting caustic dry between etch and rinse.
- Not flushing cavities; skipping the drain step.

SIGN-OFF — STATION 04

Teach-Back: why carrying caustic into acid stages causes problems; define drag-out/drag-in; what conductivity tells you and which way is “clean.”

“Rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn.”

Trainee: ____________________ Trainer: ____________________ Conductivity: ____________ µS/cm

STATION 05 OF 13

DESMUT / DEOX

"ETCHING UNCOVERS THE ALLOY'S LEFTOVERS. DESMUT WASHES THEM AWAY — AND PROTECTS YOUR COLOR."

PURPOSE & THE "WHY"

Remove the dark **smut** the etch left — the alloying elements (copper, silicon, iron, manganese) that didn't dissolve in caustic — using an **acid desmut/deox**, leaving a bright, uniform surface. Anodize over smut and you get **dark streaks, dull patches, and poor color/adhesion** — and on a color line, **uneven color.** The desmut chemistry depends on the alloy: simple alloys may need only **nitric** or **sulfuric**; high-silicon/copper alloys often need a **fluoride-bearing (or proprietary) deox.**

★ **REMEMBER**

Smut after etch is normal. Desmut's whole job is to remove it. A part that enters the anodize tank still gray/smutty comes out streaky and dull — and **colors unevenly.** Desmut is the bright-surface gatekeeper for both coating *and* color.

PARAMETER	RANGE / PRACTICE	NOTES
Chemistry	Nitric and/or sulfuric; fluoride-bearing/proprietary for high-Si/Cu	Match to the alloy
Concentration	Per product TDS	Titrate regularly
Temperature	Often ambient-warm	Per TDS
Time	~0.5–3 min	Until smut is gone, no longer
Result	Bright, uniform, active surface	No gray film remaining

⚠ **HEADS-UP — STRONG ACID (AND POSSIBLY FLUORIDE)**

Desmut is acidic; **fluoride-bearing desmuts are especially hazardous** (HF-type chemistry: deep, delayed burns, systemically toxic). Know exactly which chemistry your tank is, read its SDS, wear full acid PPE, and know the **specific first-aid** (fluoride burns may require **calcium gluconate** per your SDS/medical plan). **Add acid to water** for any make-up.

STEP-BY-STEP

1. Confirm which desmut chemistry the tank is (and its hazards/first-aid). 2. Check concentration/temp in range. 3. Immerse the specified short time — just until smut clears. 4. Withdraw, drain, move to rinse. 5. Don't over-dip — excessive desmut can etch/pit some alloys.

SIGN-OFF — STATION 05

Teach-Back: what smut is and where it came from; why anodizing over smut causes defects *and color problems*; that desmut chemistry depends on alloy; the special hazard/first-aid of a fluoride-bearing desmut.

"I removed the smut with the correct desmut for the alloy, did not over-dip, and understand the specific hazard and first-aid for this tank (including fluoride if applicable)."

Trainee: ____________________ Trainer: ____________________ Chemistry: ____________

STATION 06 OF 13

RINSE (PRE-ANODIZE)

“THE LAST CLEAN HANDOFF BEFORE THE FIRST MAIN EVENT.”

PURPOSE & THE “WHY”

Remove desmut acid before the part enters the anodize tank. Carrying desmut drag-out introduces unwanted ions (and, with a fluoride desmut, **fluoride** that can roughen/pit the anodize and spoil color uniformity). A clean rinse protects the anodize bath.

PARAMETER	RANGE	NOTES
Temperature	Ambient	—
Time	30–60 s, agitation	Longer for cavities
Water	DI preferred	Cleaner handoff to anodize
Conductivity	Per shop spec	Lower = cleaner
Flow	Counter-current overflow	—

⚠ **HEADS-UP**

Same rinse rules: drag-in acid, wet floors, goggles/gloves. If the prior tank was a **fluoride** desmut, this rinse carries fluoride — handle and dispose accordingly.

💡 **TIP**

Move from this rinse into the anodize tank **promptly and consistently.** A long, variable dwell can cause part-to-part thickness/color differences. Consistency is quality — and on a color line, consistency is *color* quality.

STEP-BY-STEP

- Confirm flow and conductivity.
- Rinse the full time with agitation.
- Drain over the tank.
- Move straight to anodize.

TOP MISTAKES

- Quick dunk instead of a real rinse.
- Inconsistent dwell before anodize.
- Ignoring conductivity.
- Forgetting fluoride drag-in handling.

SIGN-OFF — STATION 06

Teach-Back: why a clean handoff to the anodize tank matters for coating *and* color; what fluoride drag-in could do.

“The part was fully rinsed, conductivity was acceptable, and I moved promptly and consistently to the anodize tank.”

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 07 OF 13 · STEP 1, THE FIRST MAIN TANK

CLEAR SULFURIC ANODIZE — STEP 1

WHERE THE ALUMINUM'S OWN SURFACE BECOMES A HARD, POROUS CLEAR CERAMIC — THE HOME THE COLOR METAL WILL MOVE INTO

Take a breath. Up to now, every station has prepared the part. Station 07 is the **first** of this line's two main events — it grows the clear coating; Station 09 will color it.

The big idea in one sentence: **we run the aluminum part as the *anode* in cold, dilute sulfuric acid and use DC current to grow a hard, porous, CLEAR aluminum-oxide layer out of the part's own surface — to architectural Class I thickness.** No color happens here. The cast:

- **Anode** — *your aluminum part* (**positive**). Where the clear oxide grows.
- **Cathode** — **lead or aluminum (6063) plates** (negative). They carry current and bubble off hydrogen; they do **not** feed metal into the part.
- **Electrolyte** — **dilute sulfuric acid (~165–200 g/L)**, held **cold (~18–21 °C / 65–70 °F)**, with **air agitation** and **refrigerated cooling.**

★ **REMEMBER — THE ANODIZE TANK IS THE MIRROR IMAGE OF A PLATING TANK**

In plating, the part is the **cathode** and metal lands on it. Here the part is the **anode** and **oxide grows out of it.** The cathodes don't supply the coating — all the coating comes from the part's own aluminum plus oxygen from the bath. **And remember: there's no color yet. This tank makes a clear coat; the color comes in Step 2.**

⚠ **HEADS-UP — SULFURIC ACID + DC + HYDROGEN + COOLING, ALL AT ONCE**

This tank holds **strong sulfuric acid** (burns, splash, mist), runs **high-current DC** (shock/arc at contacts), bubbles **flammable hydrogen** off the cathodes, and **self-heats** (cooling must work or the coating burns). Engineering controls — **exhaust, cooling, agitation, splash control** — must be working before you run it. Full acid PPE. No ignition sources.

• WHY THE CLEAR COAT MUST BE RIGHT FOR COLOR

A uniform, well-grown, **open-pored** clear coating is the *foundation* of uniform color. A burned, powdery, or uneven clear coat colors **blotchy and off-shade** in Step 2 — and there's no fixing it with the color tank. Grow it cold, steady, and to the right thickness.

ADVANTAGE OF THE STEP-1 CLEAR COAT	WHAT IT MEANS ON THE FLOOR
Hard, integral oxide	Won't peel; protects and wears well
Porous → ready for color	The pores are the home for the color metal (Station 09)
Corrosion resistance (sealed)	Long outdoor/marine life once sealed
Class I thickness	Meets AAMA 611 / ASTM B580 architectural durability
Predictable on 6xxx/5xxx	The architectural workhorse alloys

• THE BATH MAKE-UP — KNOW THE RANGES

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES / WATCH
Sulfuric acid (H_2SO_4)	~165–200 g/L (~15–20% v/v)	The electrolyte; grows + gently dissolves the oxide
Temperature	~18–21 °C (65–70 °F)	*The* critical control — too warm = soft/powdery/burned → uneven color
Voltage (DC)	~15–18 V	Driven to hold target current density
Current density	~12–18 ASF (~1.2–2.0 A/dm²)	Sets growth rate & structure
Coating thickness	**~18–25 µm (Class I, ≥18 µm)**	Held to spec; **color is dialed in at Step 2, not here**
Time	~40–60 min (typical for Class I)	Longer = thicker, within limits
Cathodes	Lead or aluminum 6063	Carry current; bubble H_2; do not feed metal
Agitation	Air agitation	Even temperature & fresh acid at the surface
Cooling	Refrigeration / heat exchanger	Bath self-heats — *must* be cooled
Dissolved aluminum (Al^{3+})	up to ~12–15 g/L (per spec)	Some helps; too much dulls/streaks & slows the bath
Chloride contamination	Very low (≤ ~50 ppm typ.)	Chloride causes **pitting/burning**

★ **REMEMBER — THE NUMBERS TO CARRY**

Sulfuric ~165–200 g/L, temp ~18–21 °C, ~15–18 V, ~12–18 ASF, ~18–25 µm (Class I), ~40–60 min, lead/6063 cathodes, air agitation + cooling. Crucially: **on this line you grow a *consistent* clear thickness and you set the *color* later.** Don't chase color by changing the anodize thickness — chase it at the color tank.

⚙ **TECHNICAL STUFF — WHY TEMPERATURE IS KING (AND WHY IT'S A *COLOR* ISSUE HERE)**

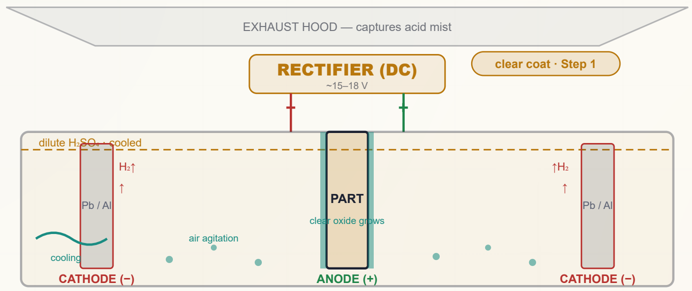
A warm bath dissolves the growing oxide too fast → a **soft, powdery, thin, or burned** clear coat. On a color line that's doubly bad: a burned clear coat has irregular pores, so it **colors patchy and off-shade.** Hold ~18–21 °C tightly and watch cooling the whole run.

⚙ **TECHNICAL STUFF — KEEP THE PORES OPEN FOR COLOR**

Everything you do after this tank — rinse, color, *then* seal — depends on the pores staying **open.** Do not let hot water or premature sealing close them before Step 2 (see Station 08). Open, clean, uniform pores = even color.

☀ **FUN FACT**

The Step-1 tank on a two-step line and an ordinary decorative Type II tank are essentially the *same tank* — same acid, same temperature, same polarity. The only difference is what happens next: instead of a dye tank, the part heads for an **AC color tank.**



Key params: `H2SO4 165-200 g/L · 18-21C · 15-18V · 12-18 ASF · 18-25um (Class I) · air agit + cooling`. Reversed polarity vs. plating — part is the ANODE (+). Clear coat only; color comes in Step 2.

STANDARD OPERATING PROCEDURE

1. **Confirm the part is clean, etched, desmutted, rinsed, racked tight** — straight from the rinse, no long air dwell.
2. **Verify engineering controls FIRST:** cooling at temperature (~18–21 °C), air agitation on, exhaust on, splash control, full acid PPE.
3. **Check bath readiness:** sulfuric in range, temp in range, dissolved-Al and chloride within limits, cathodes sound.
4. **Confirm rectifier polarity: PART = ANODE (POSITIVE).** *(Opposite of plating — verify every time.)*
5. **Set current** for the part's surface area to hit ~12–18 ASF.
6. **Calculate time** for **Class I target thickness** and set the timer/amp-hour counter.
7. **Energize and ramp** per procedure; watch **voltage rise** to hold current as the insulating oxide grows (normal).
8. **Anodize for the calculated time**, monitoring temperature, current/voltage, agitation, cooling the whole run.
9. **At target thickness, stop current, withdraw, drain over the tank**, and move to the post-anodize rinse — the pores are now **open and ready for coloring.**
10. **Log everything:** time, current/voltage, temperature, thickness checks, additions.

⚠ **HEADS-UP — VERIFY POLARITY EVERY SINGLE TIME**

Because everyone trained on plating expects the part to be the **cathode**, the most dangerous habit on this line is assuming polarity. **In Step 1 the part is the ANODE (+).** Reversed polarity gives no coating (and can damage parts/cathodes). Check before you energize.

COMMON STEP-1 DEFECTS — WHAT YOU'LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Powdery, soft, chalky coating	Bath too warm; CD too high ("burning")	Restore cooling/temp & CD; verify agitation
Thin / no coating	Bad contact; **wrong polarity**; low current; low acid	Re-rack; **verify part = anode**; check current/acid
Pitting	**Chloride contamination**; impurities	Lab-check chloride; use DI; treat bath
Dull, gray, streaky clear coat	High-Si/Cu alloy; high dissolved Al; smut not removed	Confirm alloy; lab-check Al; verify desmut
Coating that colors blotchy later	Non-uniform clear coat (temp/agitation/etch issues)	Fix the clear-coat uniformity *upstream* — the color tank can't fix it

★ **REMEMBER — A UNIFORM CLEAR COAT IS THE FOUNDATION OF UNIFORM COLOR**

Most Step-1 defects trace to **temperature/cooling** (burning), **contact** (thin/no coat), or **contamination** (pitting). All of them show up *again* as a color problem in Step 2 — and the color tank can't rescue a bad clear coat. Get Step 1 right and the color follows.

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part? *(Anode, +.)*
- Where does the clear coating come from? *(Part's own Al + O_2.)*
- The headline numbers (acid, temp, V, ASF, µm)?
- Why must the bath be cold, and what if it warms?
- Why does a uniform clear coat matter for color?
- How is thickness controlled? *(CD × time.)*
- The three safety families (acid, DC, hydrogen)?

TOP MISTAKES TO AVOID

- Assuming the part is the cathode (it's the **anode**).
- Running with failing cooling.
- Loose contact (light/burn).
- Ignoring chloride / dissolved-Al.
- Trying to set *color* by changing anodize thickness.
- Long, variable dwell before anodize.

💡 **TIP — SAMPLE BEFORE YOU COMMIT A COLOR-CRITICAL LOAD**

On any color-critical or new lot, run the **full sequence on a coupon** — clear-anodize *and* color it — and check it against your approved color standard before committing the production load. Confirming the whole recipe on the real metal is far cheaper than stripping a load.

INTEGRATED SAFETY — RESPECT THE ACID AND THE CURRENT

⚠ **HEADS-UP — SULFURIC ACID (THE HEADLINE)**
Corrosive; splash and **acid mist** hazard. Engineering controls first (exhaust, splash guards), PPE always. Any contact → flush **15 minutes**, get medical attention. **Acid to water** for make-up.

⚠ **HEADS-UP — HIGH-CURRENT DC & FLAMMABLE HYDROGEN**
LOTO for all electrical work; dry hands only; no jewelry. Hydrogen off the cathodes is **flammable** — ventilation on; no flames, sparks, or grinding nearby.

⚠ **HEADS-UP — WATCH THE COOLING THE WHOLE RUN**
The bath self-heats. If cooling fails or temperature climbs mid-run, the clear coat **burns/powders** and then **colors patchy** in Step 2. Don't set-and-forget.

★ **REMEMBER**
Step 1 is the foundation. A clean, cold, well-agitated, correctly-contacted run gives a **uniform, open-pored clear coat** — everything the color tank needs. Rush or overheat it and you've baked a color defect in before Step 2 even starts.

SIGN-OFF — STATION 07

"I can independently verify cooling/agitation/exhaust/PPE; verify bath readiness; confirm the part is racked as the ANODE (positive); set the correct current density; grow a uniform Class I clear coat by current × time; recognize burning/pitting/thin-coat defects; explain why a uniform clear coat is the foundation of uniform color; and state the acid, DC, and hydrogen hazards. Chemistry adjustments are made only by authorized personnel."

Operator: ______________________ Date: ___________ Trainer: ______________________ Date: ___________

STATION 08 OF 13

RINSE (POST-ANODIZE — KEEP THE PORES OPEN)

“GENTLE AND COOL NOW — THE PORES ARE WIDE OPEN AND WAITING FOR COLOR.”

PURPOSE & THE “WHY”

Rinse the sulfuric acid out of the freshly anodized part **without contaminating or prematurely closing the open pores**, before coloring. The fresh oxide is **porous and absorbent.** Leftover acid interferes with even coloring; impurities in the pores cause color defects. The rinse must be **clean (DI)** and **cool/ambient** — because warm water can begin to **seal the pores early and block the color metal from getting in.**

★ **REMEMBER — DON’T PRE-SEAL BY ACCIDENT**

After anodizing, **use cool/ambient DI rinse before coloring.** Hot rinse water here starts closing the pores, and a partially sealed part **won’t color properly (light, blotchy, or not at all).** Save the heat for the actual seal step — which comes *after* color.

PARAMETER	RANGE	NOTES
Water	`DI strongly preferred`	Pores absorb impurities
Temperature	`Cool / ambient (before color)`	Avoid premature sealing
Time	`1–3 min, gentle agitation`	Flush acid from pores
Conductivity	`Low (per spec)`	Confirms acid removed

⚠ **HEADS-UP**

This rinse carries dragged-out **sulfuric acid** — mildly acidic, slip hazard, goggles/gloves. Don’t let acid-laden rinse go untreated to drain.

💡 **TIP — CONSISTENCY BEFORE COLOR**

Keep the **anodize-to-color interval** consistent part-to-part. Pores age if a part sits, and aged pores color differently — a classic cause of lot-to-lot shade drift. Move from this rinse to the color tank **promptly and consistently.**

STEP-BY-STEP

- Confirm DI flow, cool temperature.
- Rinse gently, full time.
- Drain.
- Move promptly to color.

TOP MISTAKES

- Hot water before color (pre-seals pores → won’t color).
- Hard tap water (impurities → color defects).
- Short rinse leaving acid behind.
- Letting parts sit (shade drift).

SIGN-OFF — STATION 08

Teach-Back: why the pores are absorbent now; why cool DI before color; what hot water would do prematurely; why the anodize-to-color interval must be consistent.

“I rinsed with cool DI to flush acid from the open pores without pre-sealing, and moved promptly and consistently to the color tank.”

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 09 OF 13 · STEP 2, THE COLOR TANK

AC ELECTROLYTIC COLORING — STEP 2

“WHERE THE COLOR IS BORN — FINE METAL DRIVEN DEEP INTO THE PORE BOTTOMS WITH ALTERNATING CURRENT. COLOR = TIME × VOLTAGE.”

This is the station that makes this line a two-step color line. Slow down — this is the second main event, and the one that’s unique to this process.

PURPOSE & THE “WHY”

Deposit fine **metal particles (tin, nickel, or cobalt)** into the **bottoms of the open pores** of the clear-anodized part, using **AC**, to produce a durable, lightfast color from **champagne to black.** The clear coat from Step 1 is the home; here we move the color metal in. **The color is controlled at this tank — by coloring TIME and AC VOLTAGE (and metal concentration/temperature) — not by the anodize thickness.**

★ **REMEMBER — THE DEFINING STATION OF THIS MANUAL**

Step 2 = metal-salt bath + AC. On the cathodic half-cycles, metal ions are **reduced and deposited at the pore base**; the buried metal **scatters/absorbs light** for color. **Longer time / higher voltage = more metal = darker** (champagne → bronze → black). **You set the color here, then you must seal.**

⚠ **HEADS-UP — THE COLOR TANK’S THREE COMBINED HAZARDS**

This tank combines **(1) AC electrical current** in **(2) an acidic (sulfuric) solution** carrying **(3) toxic metal salts.** Treat the AC supply with the same LOTO/dry-hands discipline as the DC rectifier. The solution **burns** like any acid. And **nickel/cobalt salts are skin sensitizers and carcinogen concerns by inhalation** — run **local exhaust**, keep the solution off your skin, and never let it reach your mouth. Tin is milder but still treated as a regulated metal.

⚠ **HEADS-UP — NEVER SEAL FIRST**

Coloring only works while the pores are **open.** A part that was hot-rinsed or pre-sealed in error **won’t color.** The order is non-negotiable: **anodize → cool rinse → COLOR → rinse → seal.**

• THE BATH & PARAMETERS (TIN — THE MOST COMMON)

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES / WATCH
Stannous sulfate ($SnSO_4$)	~7–15 g/L (tin ~3–8 g/L)	The color metal source
Free sulfuric acid	~10–20 g/L	Conductivity & keeps tin in solution
Stabilizer / antioxidant additive	Per process TDS (proprietary)	Keeps tin as **Sn^{2+}** (stops Sn^{4+} sludge)
Temperature	~18–25 °C (room temp)	Affects deposition rate/shade
Current	**AC, ~10–20 V AC**	**Alternating current — the whole trick**
Time	~0.5–15 min (light → black)	**The main color knob**
Counter-electrodes	Graphite/carbon or stainless steel	Inert; complete the AC circuit
Agitation / filtration	Gentle agitation; filter	Even color; remove tin sludge

Nickel baths: nickel sulfate ~25–30 g/L + boric acid ~25–30 g/L, ~15 V AC — bronzes. Cobalt baths similar. See Part Three for tin vs. nickel/cobalt trade-offs and their greater health/waste concerns.

★

REMEMBER — TIN IS THE WORKHORSE

Tin (stannous sulfate) is the most common coloring metal — widest range, stable (with stabilizer + filtration), best deep blacks, lowest toxicity of the three. Nickel and cobalt give bronzes but bring **bigger health and waste concerns.** Whichever your shop runs, the controls are the same: **color by time & voltage, control the mist, segregate the metal waste.**

⚙

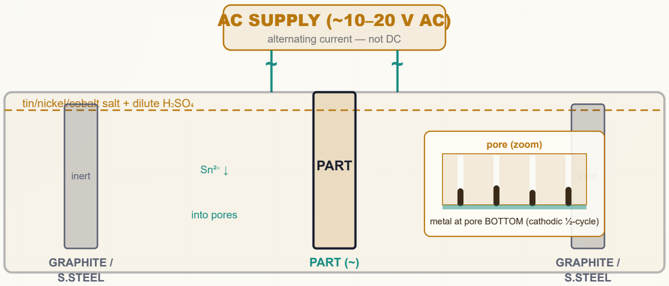
TECHNICAL STUFF — KEEP THE TIN AS SN^{2+}

A tin bath only works while the tin stays **Sn^{2+}** (stannous). Exposure to air/oxidation turns it to **Sn^{4+}**, which drops out as a useless sludge that streaks parts and clogs filters. That's why the bath carries a **stabilizer/antioxidant** and is **filtered** — and why bath maintenance (a lab/authorized-personnel job) is central to consistent color.

☀

FUN FACT

The very same tin chemistry (stannous sulfate in dilute acid) shows up in electroplating tin elsewhere in the finishing world — but here we don't plate a visible tin layer; we tuck microscopic tin specks down at the floor of each pore, where they read as color rather than as "tin."



Step 2 uses **AC.** On the cathodic half-cycle, metal ions are reduced and deposit at the pore **BASE.** Color depth = **time × voltage.** Counter-electrodes are **inert** (graphite/stainless) — they don't supply the color metal.

HOW TO GET THE COLOR RIGHT

Color depth is set by coloring time and AC voltage. Ramp the AC voltage per procedure, hold for the time that matches your target shade, and **check the color *before* sealing** — a too-light part can usually go back in for more time; a sealed wrong part must be stripped and re-anodized.

TIP — CONTROL THE VARIABLES THAT DRIFT THE SHADE

Besides time and voltage, watch **tin concentration, free acid, temperature, the stabilizer level, and the anodize-to-color interval.** Keep them consistent part-to-part and rack like-with-like. **Run to your shop's color standard and ΔE limit, not "by eye" alone.**

⚙ TECHNICAL STUFF — THROWING POWER (WHY RECESSES COME OUT LIGHTER)

Electrolytic coloring has **limited throwing power** — the AC field is weaker deep in recesses, inside corners, and channels, so those areas deposit **less metal and color lighter.** Mitigate with **electrode placement, auxiliary/conforming electrodes, agitation, and a controlled voltage ramp.** On complex extrusions, this is the #1 cause of non-uniform color — and a key reason racking and electrode setup are a real skill here.

⚙ TECHNICAL STUFF — WHY AC, RESTATED FOR THE OPERATOR

The barrier layer at the pore bottom acts as a one-way valve (rectifier). **AC** drives metal in and reduces it at the pore base on the cathodic half-cycle, while the anodic half-cycle is largely blocked from removing it — so metal **accumulates.** Plain DC would mostly just re-anodize and won't build color cleanly. **If your supply is set to DC, you will get little or no color — verify AC.**

STEP-BY-STEP

1. Confirm the part came from a **cool DI rinse with open pores** (not pre-sealed), and the anodize-to-color interval is consistent.
2. Verify the color bath: metal/acid/stabilizer in range, temperature in range, counter-electrodes sound, exhaust on, full PPE on.
3. **Confirm the supply is set to AC** at the correct voltage range. *(Not DC — verify.)*
4. Immerse fully, no trapped air (air pockets = light spots).
5. **Ramp the AC voltage** per procedure and **hold for the target time** to reach the standard shade.
6. Withdraw, drain, and **check color against the standard / ΔE** *before* sealing.
7. Too light → return for more time/voltage; on-color → rinse and move to seal.
8. Log voltage, time, temperature, color result.

★ REMEMBER — VERIFY AC, AND VERIFY COLOR BEFORE SEAL

Two habits keep this station out of trouble: **confirm the supply is AC** (DC gives no color), and **check color against the standard *before* you seal** (after sealing, you can't add more). A too-light, still-unsealed part can simply go back in the tank.

INTEGRATED SAFETY — AC, ACID, AND TOXIC METAL AT ONE TANK

⚠ HEADS-UP — AC ELECTRICAL IN A CONDUCTIVE ACID SOLUTION

Treat the AC color supply with the **same LOTO and dry-hands discipline as the DC rectifier.** Never reach into an energized tank. The solution is acidic — any contact → flush **15 minutes**, get medical attention.

⚠ HEADS-UP — TIN/NICKEL/COBALT METAL SALTS

Nickel and cobalt are skin sensitizers and carcinogen concerns by inhalation — run **local exhaust**, keep the solution off your skin, never hand-to-mouth, and wash thoroughly. Tin is milder but still a regulated metal. **Color-bath waste and rinses are a metal-salt waste stream — segregate and treat, never to drain.**

TEACH-BACK — CAN YOU ANSWER?

- What makes the color? *(Metal deposited in the pore bottoms.)*
- Which current — AC or DC? *(AC.)*
- What sets the color depth? *(Time × voltage — not anodize thickness.)*
- Which metal is most common? *(Tin.)*
- Why must the part be unsealed/open-pored?
- Why do recesses color lighter?
- The three hazards of this tank? *(AC, acid, toxic metal.)*

TOP MISTAKES TO AVOID

- Using DC instead of AC.
- Coloring a pre-sealed/aged part.
- Chasing color by changing anodize thickness.
- Sealing before checking color.
- Ignoring throwing-power in recesses.
- Breathing nickel/cobalt mist.
- Routing metal-salt waste to drain.

SIGN-OFF — STATION 09

*“I colored an open-pored clear-anodized part in the metal-salt bath under **AC**, set the shade by coloring time and voltage, verified color against the standard/ΔE **before** sealing, managed throwing-power in recesses, and handled the AC-electrical, acid, and tin/nickel/cobalt-metal hazards correctly (exhaust, PPE, waste segregation).”*

Operator: ____________________ Date: __________ Trainer: ____________________ Color/ΔE: ________

STATION 10 OF 13

RINSE (POST-COLOR)

“WASH OFF THE COLOR BATH BEFORE YOU SEAL — KEEP THE METAL SALTS OUT OF THE SEAL TANK.”

PURPOSE & THE “WHY”

Rinse **color-bath drag-out** (acid + metal salts) off the part before sealing. Carrying tin/nickel/cobalt salts into the seal tank contaminates the seal, can cause **bloom or smut**, and wastes/contaminates seal chemistry. Use **DI**, cool to ambient (the seal will supply the heat).

PARAMETER	RANGE	NOTES
Water	`DI`	Pores still partly open; keep them clean
Temperature	`Cool / ambient`	Avoid premature sealing before the seal tank
Time	`1–2 min, gentle agitation`	Flush metal-salt drag-out
Conductivity	`Low (per spec)`	Confirms color bath removed

⚠ HEADS-UP — THIS IS A REGULATED WASTE STREAM

This rinse water carries **metal salts (tin/nickel/cobalt) and acid** — it is a **regulated waste stream**, not clean rinse water. Route it to treatment, not the floor drain. Goggles/gloves; slip hazard.

💡 TIP

Don’t scrub or wipe the colored surface — the color metal is deep in the pores and durable, but a freshly colored, **unsealed** part can still pick up handling marks. Handle by the rack.

STEP-BY-STEP

- Confirm DI flow.
- Rinse gently, full time.
- Drain.
- Move promptly to seal.

TOP MISTAKES

- Skimping the rinse (metal salts into seal → bloom).
- Hot water (premature seal before color is checked).
- Wiping the unsealed colored surface.
- Sending metal-salt rinse to drain.

SIGN-OFF — STATION 10

Teach-Back: why color-bath drag-out must be rinsed before seal; that this rinse is a metal-salt waste stream.

“I rinsed the color-bath drag-out with cool DI, kept metal salts out of the seal tank, and routed the rinse to treatment.”

Trainee: ____________________ Trainer: ____________________ Date: _____________

STATION 11 OF 13

SEAL (CLOSE THE PORES)

“THE FINAL LOCK — CLOSE THE PORES TO KEEP THE COLOR METAL IN AND THE CORROSION OUT.”

PURPOSE & THE “WHY”

Close the open pores so the coating reaches full corrosion resistance and the **deposited color metal is locked in permanently.** An unsealed colored part is porous — it can absorb stains, lose performance, and corrode through the pores. **Sealing** swells/plugs the pore mouths shut, trapping the color metal deep inside. The classic method is **hot DI water near boiling** (forms swollen **boehmite**); there are also **mid-temperature nickel-acetate** and **cold/room-temperature nickel-fluoride** seals.

★ **REMEMBER — SEALING IS THE POINT OF NO RETURN**

After sealing you **cannot add color** (pores are shut). Seal is the *last* chemical step. **Verify color BEFORE you seal.** A poorly sealed part will weather and stain; an over-sealed/contaminated part can show **“sealing bloom”** (a chalky film that dulls a dark bronze). Sealing finishes a beautiful job — or quietly ruins it.

THE THREE COMMON SEALS

SEAL TYPE	TEMPERATURE	TRADEOFFS
Hot DI water seal	~96–100 °C	Simple, metal-free, excellent quality; high energy, slow (~2–3 min/µm), bloom-prone if water is off
Nickel-acetate (mid-temp)	~80–90 °C	Lower energy; excellent corrosion & color retention; widely used architecturally; contains **nickel**
Cold nickel-fluoride	~25–30 °C	Lowest energy/fastest; often a warm-water “aging” rinse after; contains **nickel + fluoride**; needs tight control

⚙ **TECHNICAL STUFF — WHY HOT WATER SEALS**

Near-boiling water hydrates the pore-wall oxide to **boehmite (AlOOH)**, which occupies more volume than the original oxide and **swells the pore mouths shut** — locking the color metal in the pore bottom. Nickel seals add nickel salts that precipitate in the pore mouths to help plug them, which is why they work at lower temperature.

⚙ **TECHNICAL STUFF — SEALING BLOOM ON A COLORED PART**

A powdery white/gray film after sealing (“bloom”) usually comes from **hard water, drag-in, wrong pH, or over-concentrated seal.** On a two-step bronze or black, bloom is **especially visible** — it dulls and grays the color. The fix is **clean DI water and correct seal chemistry/pH/temperature**, plus a light post-dip if needed.

⚠ HEADS-UP — NEAR-BOILING TANK = SCALD HAZARD

A hot-water or hot nickel seal runs **~80–100 °C.** Splashes and steam cause **serious scald burns.** Face shield, heat-resistant gloves, long sleeves; mind the steam. Lower parts gently.

⚠ HEADS-UP — NICKEL SEALS ARE A HEALTH & WASTE CONCERN

Nickel is a skin sensitizer, and the **fluoride** in cold seals is especially hazardous (deep, delayed burns; systemic toxicity). Gloves and goggles, read the SDS, know the fluoride first-aid (**calcium gluconate** per your plan), and route nickel/fluoride seal waste to proper treatment — **never to drain.** *(Note: between the color bath and the seal, this line can put nickel into the waste stream from two sources — color and seal — so segregation matters even more.)*

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Water	`DI (hot-water seal)`	Hard water → bloom
Temperature	`Hot ~96–100 · Ni-acetate ~80–90 · Ni-F ~25–30 °C`	Per chosen seal
pH	`Per seal product (often ~5.5–6.5)`	Wrong pH = poor seal/bloom
Time	`~2–3 min per µm (hot water)`	Per seal type & thickness
Result	Closed pores; passes seal-quality test	Dye-stain or admittance test (Part Three)

STEP-BY-STEP

1. Confirm seal type, temperature, pH, DI water in range. 2. Confirm the part came from a **clean post-color rinse** and **color was verified.** 3. Immerse gently (mind the scald), full time for the thickness. 4. Withdraw, rinse if specified, inspect for **bloom.** 5. Move to final rinse/dry.

TEACH-BACK — CAN YOU ANSWER?

- What sealing does (closes pores) and why (lock color metal + corrosion).
- Why you can't color after sealing.
- The three seal types and one tradeoff of each.
- The scald hazard; the nickel/fluoride hazard.
- What sealing bloom is and its usual cause.

TOP MISTAKES

- Sealing before color is verified.
- Wrong temp/pH → incomplete seal (weathers/corrodes).
- Hard/tap water → bloom.
- Plunging into a near-boiling tank.
- Routing nickel/fluoride waste to drain.
- Skipping the seal-quality check.

SIGN-OFF — STATION 11

"I verified color before sealing, sealed at the correct type/temperature/pH in DI water, checked for bloom, and handled the scald and (if applicable) nickel/fluoride hazards correctly."

Trainee: ____________________ Trainer: ____________________ Seal type/temp: ____________

STATION 12 OF 13

RINSE / DRY

“FINISH CLEAN, DRY GENTLY, DON’T UNDO YOUR WORK.”

PURPOSE & THE “WHY”

Give the sealed, colored part a final clean rinse and a **gentle dry** without water spotting or thermal damage. A clean final rinse removes seal-tank drag-out so the part dries spot-free; drying should be **gentle** — a sealed, dark-colored part shows water spots and handling marks badly at the very end.

PARAMETER	RANGE	NOTES
Final rinse water	`DI`	Spot-free drying
Rinse temperature	`Warm`	Aids drying
Dry method	`Air / warm forced air; gentle oven`	Avoid excessive heat
Handling	`By rack / clean gloves`	No bare-hand prints on finished parts

⚠ **HEADS-UP**

Parts off a hot seal are **hot** — burn risk. Let them cool or use gloves. Dry ovens are a heat hazard; mind hot racks.

💡 **TIP**

A **DI final rinse** is the cheapest insurance against water spots on a finished dark-bronze or black part — tap-water spots on those colors are glaringly obvious and pure rework.

STEP-BY-STEP

- Final DI rinse.
- Drain.
- Gentle dry.
- Handle by rack/gloves to inspection.

TOP MISTAKES

- Tap-water final rinse (spots).
- Over-aggressive drying.
- Bare-hand handling of finished colored parts.

SIGN-OFF — STATION 12

Teach-Back: why a clean final rinse + gentle dry protects the finished color.

“I gave a clean DI final rinse, dried gently without spotting/overheating, and handled finished parts cleanly.”

Trainee: ____________________ Trainer: ____________________ Date: _____________

STATION 13 OF 13

FINAL INSPECTION

"PROVE IT'S RIGHT — COLOR/ΔE, THICKNESS, SEAL, AND FINISH."

PURPOSE & THE "WHY"

Verify the finished part meets spec: **color match (ΔE), coating thickness, seal quality, and appearance** — to AAMA 611 / project spec. On a color line, **color is the headline check**: compare to the approved standard, measure **ΔE** with a spectrophotometer where required, and confirm uniformity (no streaks, no light recesses). **Thickness** (eddy-current) confirms the Class I clear coat; **seal quality** (dye-stain or admittance) confirms the pores closed (locking the color in); appearance catches bloom, burns, light spots, and handling damage.

CHECK	METHOD	PASS CRITERIA
Color / ΔE	Visual vs. standard; **spectrophotometer (ΔE)**	Within ΔE limit; uniform (incl. recesses) per AAMA 611/project
Coating thickness	**Eddy-current gauge** (ASTM B244) on significant surfaces	Class I (≥18 μm) per spec
Seal quality	**Dye-stain** (ASTM B136) and/or **admittance** (ASTM B457)	Low dye pickup / low admittance = well sealed
Lightfastness/weathering	Per AAMA 611 / project (lab or certified data)	Meets the spec's weathering requirement
Appearance	Visual under controlled light	No bloom, burns, streaks, light recesses, contact marks
Dimensions	Gauging where critical	Within tolerance **after** dimensional growth

★

REMEMBER — THE CHECKS THAT MATTER MOST HERE

Color/ΔE = "right shade and uniform?", **eddy-current = "enough clear oxide grown?"**, and the **seal test = "pores closed (color locked)?"** A part can be the right color and thickness and *still fail* if it isn't sealed — which is why the seal test exists, and why color is checked *before* sealing but confirmed *after.*

💡

TIP — THE DYE-STAIN SEAL TEST IN PLAIN TERMS

Drop a spot of test dye on a sealed area; a **good seal barely picks up color** (wipes clean); a **poor seal stains** (pores still open). It's the quick visual cousin of the more precise **admittance** test.

SIGN-OFF — STATION 13

Teach-Back: which check answers "right color/uniform?" vs. "enough oxide?" vs. "pores closed?"; what a dye-stain result means; why recesses get extra color scrutiny.

"I verified color/ΔE against the standard including recesses, measured thickness on significant surfaces, ran the seal test, inspected for defects, and confirmed critical dimensions accounting for growth."

Trainee: ____________ Trainer: ____________ ΔE: ______ · Thickness: ______ μm · Seal: P/F

PART THREE — ACROSS THE LINE
DEEP DIVE

COLOR CONTROL BY COLORING TIME & VOLTAGE

THE IDEAS THAT DON'T LIVE IN ONE TANK — THEY TIE THE WHOLE LINE TOGETHER

On a two-step line, **color is set at the color tank — by coloring TIME and AC VOLTAGE — not by the anodize thickness.** This is the single most important operating contrast with integral color.

In integral color, "darker" means "thicker coating on a tightly controlled alloy." Here, you grow a **consistent Class I clear coat** and then dial the shade entirely in Step 2: a short, low-voltage dip gives champagne; a longer dip at higher voltage gives black. Secondary variables that drift the shade — **tin (or Ni/Co) concentration, free acid, bath temperature, stabilizer level, and the anodize-to-color interval** — must be held consistent. **Always run to a color standard and ΔE limit**, and **verify color before sealing** so a too-light part can return to the color tank.

LEVER	EFFECT ON COLOR	OPERATOR ACTION
Coloring time	More = more metal = darker	Hold to the time for the target shade
AC voltage	Higher = faster/darker deposition	Ramp per procedure; don't overshoot
Metal concentration / free acid	Drift the rate & shade	Maintain per TDS (lab/authorized)
Bath temperature	Shifts deposition rate	Hold steady; run like-with-like
Anodize-to-color interval	Aged pores color differently	Keep the interval consistent
Clear-coat thickness	Held to spec — **not** the color knob	Grow consistent Class I; set color at Step 2

★ **REMEMBER**
Decoupling color from coating thickness is *the* advantage. It means a fixed, spec-driven clear coat for **durability** *and* an independently controlled color for **appearance** — far more repeatable than "color = thickness."

💡 **TIP**
When a shade drifts lot-to-lot, don't reach for the anodize tank. Check the **Step-2 variables first** — time, voltage, metal/acid concentration, temperature, stabilizer, and the anodize-to-color interval — and confirm the clear coat was uniform. The color knob is the color tank.

☀ **FUN FACT**
Because color and thickness are independent here, two parts can be the *same* dark bronze with slightly different clear-coat thicknesses, or the *same* thickness in two different shades — something integral color simply can't do, since for it thickness **is** the color.

DEEP DIVE

THE AC COLORING MECHANISM — WHY ALTERNATING CURRENT

WHY THE COLOR STEP USES AC AND NOT DC — THE BARRIER LAYER IS A ONE-WAY VALVE

The dense **barrier layer** at the pore bottom behaves like a **rectifier (a one-way electrical valve).** Under plain **DC**, it mostly keeps re-anodizing and won't let you build metal cleanly. **AC** exploits the asymmetry:

- On the **cathodic half-cycle**, metal cations (Sn^{2+}, Ni^{2+}, Co^{2+}) are pulled down the pores and **reduced to solid metal at the pore base.**
- On the **anodic half-cycle**, the barrier layer largely **blocks the metal from redissolving.**
- Cycle after cycle, **metal accumulates into columns at the pore bottom**, and the amount of accumulated metal sets the color (more metal = darker).

The metal colors by **optical absorption and scattering** of light entering the coating. Modern processes may use **shaped/asymmetric or pulsed AC waveforms** for finer control — read your process TDS. **Throwing power is limited**, so recesses color lighter and electrode setup/agitation matter.

★ REMEMBER

AC works because the barrier layer is a one-way valve. The cathodic half-cycle deposits metal at the pore base; the anodic half-cycle can't pull it back out — so it builds up. **DC would mostly re-anodize and give little or no color — verify the supply is AC.**

⚙ TECHNICAL STUFF — THROWING POWER, RESTATED

The AC field is weaker deep in recesses, inside corners, and channels, so those areas deposit **less metal and color lighter.** On complex extrusions this is the #1 cause of non-uniform color. Mitigate with **electrode placement, auxiliary/conforming electrodes, agitation, and a controlled voltage ramp.**

⚙ TECHNICAL STUFF — THE COLOR IS OPTICAL, NOT PIGMENT

The buried metal particles don't "contain" a color the way a pigment does — they **absorb and scatter** incoming light, and the apparent shade depends on how much metal is there and how it's distributed in the pore. That's why "more metal = darker" runs smoothly from champagne through bronze to black.

☀ FUN FACT

The same rectifying behavior of the barrier layer that makes ordinary DC *useless* for coloring is exactly what makes **AC** work so well — the process turns a nuisance (a one-way valve) into the very mechanism that traps the color metal in place.

DEEP DIVE

WHY TWO-STEP BEATS INTEGRAL COLOR

BOTH ARE DURABLE AND LIGHTFAST — BUT TWO-STEP WINS ON WHAT PRODUCTION CARES ABOUT

Both routes give durable, lightfast bronzes and blacks — but two-step wins on the things production cares about:

- **Energy.** Integral color is a thick, high-voltage, heavily-cooled “self-coloring” anodize. Two-step keeps Step 1 as an ordinary Type II anodize and adds a **short, low-voltage AC color dip** — **less total energy.**
- **Consistency / repeatability.** Integral color’s color *is* the alloy, so heats/extrusions/tempers shift the shade. Two-step **deposits the color itself**, so it’s **far less alloy-sensitive** — **lot-to-lot matching is dramatically easier.**
- **Tonal range.** Two-step (especially tin) gives a **wider range and superior deep blacks.**

The trade is two process steps and a second tank to run and maintain (filtration, stabilizer, metal-salt waste). For most modern architectural work, that trade is worth it — which is why two-step has **largely displaced integral color** for new construction. Integral color is still specified for legacy matching and its proven track record.

PRODUCTION CONCERN	INTEGRAL COLOR	TWO-STEP ELECTROLYTIC
Energy	High (high V, long cycle, big cooling)	**Moderate (short low-V AC dip)**
Lot-to-lot consistency	Hard (alloy-driven)	**Very good (you deposit it)**
Alloy sensitivity	High	**Low**
Deep blacks / tonal range	Limited	**Excellent (tin)**
Tanks to run	One (color is in the anodize)	Two (clear anodize + color)
Best fit today	Legacy match / proven spec	**New architectural color**

★ **REMEMBER**

Two-step matches integral color’s **durability and lightfastness** while winning on **energy, consistency, and blacks** — at the cost of a second tank. That balance is exactly why it’s the modern architectural default.

☀ **FUN FACT**

An architect can re-order the *same* dark bronze year after year from a two-step line and get a consistent match, even as the supplied aluminum comes from different mills and heats — the kind of repeatability that was a constant headache in the integral-color era.

REFERENCE

THE THREE COLORING ROUTES COMPARED

WHEN TO CHOOSE TWO-STEP — AND WHEN TO POINT THE CUSTOMER ELSEWHERE

	DYE (TYPE II)	INTEGRAL COLOR	TWO-STEP ELECTROLYTIC
Color source	Organic dye in pores	Alloy + oxide structure	**Metal in pore bottoms (AC)**
Lightfastness	Low–medium	Excellent	**Excellent**
Palette	Any color	Champagne→bronze→black	Champagne→bronze→**black**
Energy	Low	**High**	**Moderate**
Consistency	Good	**Hard (alloy-driven)**	**Very good**
Alloy sensitivity	Low–moderate	**High**	**Low**
Best for	Indoor/consumer	Exterior (legacy/spec)	**Exterior (modern std)**

★ **REMEMBER — ROUTE TO REQUIREMENT**

Bright color indoors → dyed Type II. Durable bronze/black outdoors → integral or two-step. When cost, energy, lot-to-lot consistency, and deep blacks matter — which is most modern architectural work — **two-step wins.** Two-step can't make a bright primary color; for that, steer the customer to dyed Type II (interior).

💡 **TIP — READING A SPEC**

"Dark bronze / black, AAMA 611, Class I, no dye" usually means **two-step electrolytic** (or legacy integral). "Bright red/blue/gold, interior" means **dyed Type II.** When in doubt, confirm the spec and ask — the chemistry, cost, and durability all change with the route.

☀ **FUN FACT**

All three routes can make a bronze that looks identical on day one. Ten years in the sun sorts them out: the **dyed** one fades, while the **integral** and **two-step** bronzes hold — durable ceramic/metal color, not an organic dye.

DEEP DIVE

TIN VS. NICKEL VS. COBALT COLORING BATHS

WHICH METAL, WHICH COLOR RANGE — AND WHICH CARRIES THE BIGGER HEALTH/WASTE BURDEN

BATH	TYPICAL CHEMISTRY	COLOR RANGE	NOTES & HAZARDS
Tin (most common)	Stannous sulfate + H_2SO_4 + stabilizer	Champagne → bronze → **black**	Widest range, stable (with stabilizer/filtration), cost-effective; the architectural default. Tin is comparatively low-tox but still a treated metal.
Nickel	Nickel sulfate + boric acid + additives	Bronzes	Good bronzes; **nickel is a skin sensitizer and inhalation carcinogen concern** — control mist; regulated waste.
Cobalt	Cobalt sulfate + additives	Bronzes / black	Less common; **cobalt is a sensitizer/carcinogen concern** — control mist; regulated waste.
Tin-nickel (mixed)	$SnSO_4$ + $NiSO_4$ blends	Extended range	Used for specific shades; carries nickel concerns.

★ **REMEMBER**

Tin is the workhorse — good range, stable, best blacks, lowest toxicity of the three. Nickel/cobalt give bronzes but bring **bigger health and waste concerns.** Whichever your shop runs, control the **mist** and **segregate the metal waste.**

⚠ **HEADS-UP — NICKEL & COBALT MIST**

Nickel and cobalt salts are skin sensitizers and inhalation carcinogen concerns. Local exhaust at the tank is your first defense; gloves and goggles keep them off skin and eyes; no hand-to-mouth. A tin bath is milder, but the same exhaust/PPE discipline keeps any acid mist out of your lungs.

⚙ **TECHNICAL STUFF — WHY TIN NEEDS A STABILIZER**

Tin works only while it stays **Sn^{2+}**; air oxidizes it to **Sn^{4+}**, which precipitates as sludge that streaks parts and clogs filters. The bath therefore carries a **stabilizer/antioxidant** and is filtered. Nickel/cobalt baths (sulfate + boric acid) are more stable in that respect but carry the bigger toxicity/waste load.

☀ **FUN FACT**

The famous deep architectural **black** is overwhelmingly a **tin** color — tin's tonal range runs cleanly all the way to a rich black, which is a big reason it became the industry default over the nickel/cobalt bronzes.

DEEP DIVE

RACKING, SEAL, RINSING & WATER/DI

THE CROSS-CUTTING PRACTICES THAT PROTECT COLOR, DURABILITY, AND FINISH

• RACKING

The part is the anode in Step 1 and carries AC in Step 2, so **the rack is the wire for both coating and color.** Good racking = firm contact, hidden contact point, orientation for gas/drainage, no parts touching, maintained (titanium) racks — *and* electrode/auxiliary setup that gives **even color into recesses.** *(Cross-reference Stations 01 and 09.)*

• SEAL

Seal closes the pores and **locks the color metal in**: **hot DI (~96–100 °C)** — metal-free, high energy, bloom-prone; **nickel-acetate (~80–90 °C)** — efficient, excellent retention, contains nickel; **cold nickel-fluoride (~25–30 °C)** — fast/low-energy, nickel + fluoride hazard. Confirm the seal with **dye-stain (ASTM B136)** or **admittance (ASTM B457).**

• RINSING & WATER

Use **cool DI** for the post-anodize and post-color rinses to keep pores open and clean; tap-water impurities (chloride, hardness) cause **pitting, color defects, and bloom.** Counter-current cascades save water; conductivity gauges cleanliness. Treat **DI water** as a color-quality tool, not a utility — the chloride that pits the anodize and the hardness that blooms the seal are invisible in tap water.

> ⚠ **HEADS-UP**
> Acidic and caustic streams must be **neutralized under control** (never casually mixed). **Color-bath rinses (tin/nickel/cobalt) and nickel/fluoride seal waste** need **dedicated treatment** — never to drain. Know your shop's segregation scheme.

> ★ **REMEMBER**
> Three cross-cutting habits protect every two-step job: **tight, like-with-like racking with good electrode setup** (color & contact), **a clean, correct seal** (locks color, no bloom), and **cool DI rinsing** (open pores, no pits, no spots). They sound minor — they decide whether a beautiful color survives.

> 💡 **TIP**
> On complex extrusions, plan the **electrode/auxiliary setup** at the color tank the way you'd plan racking — it's the difference between uniform color and light recesses. Throwing power is a racking-and-electrode skill, not just a bath setting.

THE TIN/NICKEL/COBALT METAL LOAD

WASTE & ENVIRONMENTAL

SEVERAL REGULATED STREAMS — NEVER TO A FLOOR DRAIN UNTREATED

A two-step line generates several regulated streams — **never to a floor drain untreated:**

- **Spent sulfuric & acid rinses** — neutralize/treat; manage dissolved aluminum (aluminum hydroxide sludge).
- **Spent caustic etch** — high pH, dissolved aluminum; neutralize/treat.
- **Desmut waste** — acidic; **fluoride-bearing** desmuts need fluoride treatment (precipitate as calcium fluoride).
- **Color-bath waste & post-color rinses — tin / nickel / cobalt metals + acid. Nickel and cobalt are regulated, toxic metals** (and the bath carries tin sludge); segregate, precipitate metals, and treat. **Never to sewer.**
- **Nickel-seal waste** (nickel-acetate / nickel-fluoride) — **nickel regulated**; fluoride needs treatment; segregate.
- **Air** — sulfuric acid mist, near-boiling steam, and **nickel/cobalt-bearing color-bath mist** need exhaust/scrubbing per permit; **avoid worker inhalation of color-bath mist.**

⚠ HEADS-UP

This line can put **nickel into the waste from two places** — the color bath (if nickel/cobalt) *and* a nickel seal. Acid and caustic streams must be **neutralized under control** (never casually mixed). Know your shop's segregation scheme; the wrong drum or drain can be a reportable violation.

★ REMEMBER

The streams unique to a two-step line are the **metal-salt color-bath waste and rinses (tin/nickel/cobalt)** and any **nickel/fluoride seal** waste — on top of the usual acid/caustic neutralization and dissolved-aluminum sludge. **Segregate and treat every metal-bearing stream by your permit.**

⚙ TECHNICAL STUFF — PRECIPITATING THE METALS OUT

Metal-bearing streams are typically treated by **pH adjustment to precipitate the metals as hydroxides** (and fluoride as calcium fluoride), then clarified and filtered, with the metal sludge handled as regulated waste. The clarified water is discharged only within permit limits. This is why color-bath rinses can't go straight to drain — the metals have to come out first.

☀ FUN FACT

Two-step color avoids the hexavalent-chromium chemistry of some old color routes — a genuine hazard reduction — but it trades that for a **tin/nickel/cobalt metal load** in the waste stream. Different burden, same rule: treat it, don't dump it.

HOW WE PROVE IT

QUALITY CHECKS & TEST METHODS

THE MAKE-OR-BREAK CHECKS — COLOR/ΔE, LIGHTFASTNESS, THICKNESS, AND SEAL

CHECK	METHOD (TYPICAL STD)	ANSWERS
Color / ΔE	Spectrophotometer / visual vs. standard	“Right shade & uniform?”
Lightfastness / weathering	Accelerated UV/weatherometer or natural exposure per **AAMA 611**	“Will the color survive service?”
Coating thickness	Eddy-current (ASTM B244); micro-section if needed	“Enough clear oxide (Class I)?”
Seal quality	Dye-stain (ASTM B136); admittance (ASTM B457); acid-dissolution weight loss	“Pores closed (color locked)?”
Abrasion / wear	Per spec	“Durable enough?”
Corrosion	Salt spray (ASTM B117) per spec	“Survives service?”
Appearance/defects	Visual under controlled light	Bloom, burns, streaks, light recesses, marks

★ **REMEMBER**

Architectural two-step color is typically certified to **AAMA 611** and **ASTM B580 (architectural Type A)** with **Class I = ≥18 µm (0.7 mil)** for exterior durability. The make-or-break field checks are **color/ΔE, thickness, and seal** — with lightfastness/weathering proven by spec data.

💡 **TIP — MEASURE COLOR LIKE A PRO**

Read ΔE on the **significant surface** under **controlled light**, against the **approved standard** — not “by eye” under shop lighting and not part-to-part. Architectural specs set a maximum ΔE and a light/dark range; keep **retained range samples** to judge against, and check **recesses** (throwing power leaves them lighter).

⚙ **TECHNICAL STUFF — THICKNESS, SEAL, AND COLOR TOGETHER**

No single number proves the part. **Thickness** says you grew enough clear oxide; the **seal test** says the pores are closed (color locked in); and **color/ΔE** says the shade is right and uniform. A part can pass thickness and still fail color (wrong time/voltage) — which is why color is non-negotiable on this line.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

DEFECT / SYMPTOM	LIKELY CAUSE	FIX
Color too light / won't go dark	Too little coloring time/voltage; low metal conc.; aged pores; pre-sealed	Increase color time/voltage; check bath metal/stabilizer; color promptly; never pre-seal
Color too dark	Too much coloring time/voltage	Reduce time/voltage; catch *before* seal
Non-uniform color / light recesses	Throwing-power; poor electrode setup; shadowing; uneven clear coat	Improve electrode/auxiliary setup & agitation; rerack; fix clear-coat uniformity
Color streaks	Tin sludge (Sn^{4+}); contamination; uneven etch/desmut/clear coat; drag-in	Filter/maintain bath & stabilizer; verify clean/etch/desmut; consistent dwell
Color mismatch lot-to-lot	Inconsistent time/voltage/temp/conc.; anodize-to-color interval drift; mixed alloys	Standardize Step-2 settings & interval; rack like-with-like; run to ΔE standard
No color at all	Pre-sealed part; DC instead of AC; no contact; dead bath	Verify open pores; **verify AC supply**; re-rack; lab-check bath
Burned/powdery clear coat	Step-1 bath too warm; CD too high	Restore cooling/agitation; lower CD
Pitting	Chloride contamination (Step 1)	Lab-check chloride; use DI; treat bath
Thin/no clear coat	Poor contact; **wrong polarity (Step 1)**; low current/acid	Re-rack; **verify part = anode**; check current/acid
Poor seal (fails test)	Wrong seal temp/pH; contaminated seal; metal-salt drag-in	Correct temp/pH/DI; rinse color drag-out before seal; re-seal
Sealing bloom (chalky film)	Hard water; wrong pH; over-concentrated/contaminated seal	DI water; correct pH/conc.; light post-dip

TIP

Most two-step *color* defects trace to one of four roots: **the clear coat** (uneven Step 1 → uneven color), **the color bath** (time/voltage/metal/sludge), **throwing power** (recesses light), or **order/handling** (pre-sealed, DC-not-AC, drag-in, aged pores). Name the root, not just the symptom.

★ REMEMBER

Two of these are **safety-relevant** as well as quality: **wrong polarity in Step 1** (verify part = anode) and **DC instead of AC in Step 2** (verify the color supply). When in doubt about a bath change, ask your supervisor before you act.

EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

PLAIN-LANGUAGE DEFINITIONS FOR EVERYTHING IN THIS MANUAL

AC (alternating current): current that reverses direction many times a second; used in **Step 2** to deposit color metal in the pore bottoms.
Admittance test: electrical test (ASTM B457) of how “open” the coating still is — confirms sealing (lower = better seal).
Anode: the **positive** electrode. In Step-1 anodizing it’s **your part** — where the clear oxide grows.
Anodizing: growing an integral oxide layer out of a metal’s surface by making the part the anode in an electrolyte.
Barrier layer: the thin, dense oxide at the pore bottom (against the metal); acts as a **rectifier**, which is why AC is used for coloring.
Boehmite: hydrated aluminum oxide (AlOOH) formed during hot-water sealing; its swelling closes the pores.
Cathode: the **negative** electrode (lead/aluminum in Step 1) — carries current and bubbles off hydrogen; does **not** feed the coating.
Class I: architectural-grade anodic coating, **≥18 µm (0.7 mil)** — the usual two-step clear-coat spec.
ΔE (Delta-E): a number for how different two colors are; used to set/pass color tolerances.
Electrolytic coloring (electrocoloring): depositing metal in the pore bottoms with **AC** — Step 2 of this process.
Integral color: color made *by the anodizing itself* (alloy + organic-acid oxide); durable but energy-intensive and alloy-sensitive — the route two-step largely replaced.
Lightfastness: resistance of a color to fading under UV/weather; two-step colors are excellent.
Porous layer: the thick upper oxide full of vertical **pores**; in two-step, the **bottoms** of the pores hold the color metal.
Sealing: the final step that **closes the pores** (hot water, nickel-acetate, or cold nickel-fluoride), locking in the color metal and corrosion resistance.
Stannous sulfate ($SnSO_4$): the tin salt in the most common coloring bath; the tin must stay **Sn^{2+}** (a stabilizer prevents Sn^{4+} sludge).
Throwing power: how evenly the color deposits into recesses; limited in electrolytic coloring (recesses color lighter).
Two-step electrolytic color: clear anodize (Step 1) + AC metal coloring in the pore bottoms (Step 2) — the modern architectural color standard.
Valve metals: metals that anodize to protective oxides (aluminum, titanium, magnesium, tantalum, niobium).
Water-break test: a free cleanliness check — clean metal sheets water; dirty metal beads it.

★ **REMEMBER**

Five terms matter most: **anode** (your part in Step 1), **AC coloring** (Step 2 — metal in the pore bottoms), **color = time × voltage** (not thickness), **Class I clear coat** (the durable foundation), and **seal** (locks the color metal in). Master those and you understand two-step color.

☀ **FUN FACT**

The word **“anode”** comes from Greek *ánodos*, “a way up” (Faraday’s term). On this line there’s a nice symmetry: the **anode** is the “way up” where your clear coating grows, and then **alternating current** sends color metal the “way *down*” to the very bottom of the pores.

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

Sign-off levels: **O** = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	**Safety / PPE / SDS / hygiene — acid, caustic, hot seal, hydrogen, DC, AC, tin/nickel/cobalt metals (all stations)**	____	______	______	_____
2	Station 01 — Inspection & Racking (contact)	____	______	______	_____
3	Station 02 — Clean / Degrease + water-break	____	______	______	_____
4	Station 03 — Etch (caustic, architectural satin)	____	______	______	_____
5	Station 04 — Rinse + conductivity	____	______	______	_____
6	Station 05 — Desmut / Deox	____	______	______	_____
7	Station 06 — Rinse	____	______	______	_____
8	**Station 07 — Clear Anodize (Step 1, part = anode)**	____	______	______	_____
9	Clear-coat thickness control (CD × time) & dimensional growth	____	______	______	_____
10	Station 08 — Post-anodize rinse (keep pores open)	____	______	______	_____
11	**Station 09 — AC Electrolytic Coloring (Step 2)**	____	______	______	_____
12	Color control (time × voltage), throwing-power, ΔE-to-standard	____	______	______	_____
13	Station 10 — Post-color rinse (metal-salt drag-out)	____	______	______	_____
14	Station 11 — Seal (close pores, lock color)	____	______	______	_____
15	Station 12 — Rinse / Dry	____	______	______	_____
16	Station 13 — Final Inspection (color/ΔE, thickness, seal)	____	______	______	_____
17	LOTO on **DC rectifier AND AC color supply** / electrical safety	____	______	______	_____
18	Emergency response — eyewash, shower, spill, fluoride first-aid	____	______	______	_____
19	Waste segregation & environmental (incl. **tin/nickel/cobalt metals**)	____	______	______	_____
20	Troubleshooting — color & coating symptom recognition	____	______	______	_____

Operator name: ______________________ **Hire/start date:** ______________

Supervisor name: ______________________ **Review cycle:** ☐ Annual ☐ Other: __________

⚠ **HEADS-UP**

Safety rows (1, 17, 18, 19) are not optional and not “fill in later.” An operator may not work *any* station until PPE/SDS/hygiene, **LOTO (both the DC and AC supplies)**, emergency response (including **fluoride first-aid** where used), and waste-control awareness (**including tin/nickel/cobalt metals**) are signed. Safety competency precedes production competency.

PART FOUR — CERTIFICATION
20 QUESTIONS · PASSING SCORE 80% (16/20)

COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**
Passing score: 80% (16 of 20). Any **safety** question answered wrong (**Q3, Q9, Q13, Q17, Q19**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ____________________ **Date:** ____________ **Score:** ________ / 20 **Safety-gate Qs: 3, 9, 13, 17, 19**

Q1. On a two-step electrolytic color line, the color comes from:
A. Organic dye soaked into the pores after anodizing B. The alloy itself during anodizing C. Fine metal (tin/nickel/cobalt) deposited deep in the bottom of the pores using AC, after a clear anodize D. Pigment added to the seal tank

Q2. Anodizing differs from electroplating mainly because in anodizing the part is the:
A. Anode, and an oxide is grown out of the part itself B. Cathode, and a different metal is deposited on it C. Insulator, and nothing happens D. Cathode, and it dissolves

Q3. [SAFETY GATE] Beyond the acids/caustic of Step 1, the Step-2 coloring tank adds which hazards?
A. Radiation and noise B. None — it's just water C. Only extra lighting D. AC electrical current in an acidic metal-salt bath, plus toxic metal salts (nickel/cobalt are sensitizers/carcinogen concerns by inhalation) and their mist

Q4. On this line, the color is made **darker** by:
A. Growing a thicker clear anodize B. Coloring longer and/or at higher AC voltage (depositing more metal) C. Using a stronger dye D. Sealing longer

Q5. The two steps, in order, are:
A. Color, then anodize B. Seal, then color C. (Step 1) a clear sulfuric (Type II) anodize, then (Step 2) AC electrolytic coloring in a metal-salt bath D. Dye, then integral color

Q6. (Short answer) Explain in your own words why anodizing is **not** plating — name which electrode the part is in Step 1 and **where the clear coating comes from.**

Q7. The most common electrolytic coloring metal/bath is:
A. Tin (stannous sulfate) B. Gold C. Chromium D. Zinc

Q8. The biggest reason two-step color has **largely replaced integral color** for architecture is:
A. It's the only durable option that exists B. It's more energy-efficient and far more consistent/repeatable across alloys and lots (color decoupled from thickness/alloy), with excellent blacks C. It uses no electricity D. It needs no clear anodize

Q9. (Short answer) [SAFETY GATE] Name the **two corrosive hazards** on the line (one high-pH, one low-pH), and state the universal rule for adding acid during bath make-up.

Q10. Step 2 (coloring) uses which kind of current?
A. High-voltage DC only B. No current at all C. Microwave energy D. AC (alternating current)

Q11. Compared with integral color, two-step color depth is controlled by ____ rather than coating thickness/alloy:
A. The seal temperature B. The dye concentration C. The Step-2 coloring time and AC voltage D. The etch time

Q12. (Short answer) What is **smut**, where does it come from, and which station removes it? Why does a uniform clear anodize matter for even **color** on this line?

Q13. [SAFETY GATE] The metals in nickel/cobalt coloring baths are a special concern because:
A. They are flammable   B. They are radioactive   C. Nickel and cobalt are skin sensitizers and carcinogen concerns by inhalation — avoid mist/aerosol and skin contact   D. They are completely harmless

Q14. A part comes out of the color tank too **light**. The correct response is:
A. Catch it before sealing — return it to the color tank for more time/voltage to reach the standard (and if it's already sealed, strip and re-anodize)   B. Dip it in dye to darken it   C. Paint it darker   D. Seal it twice

Q15. Architectural two-step color is typically specified at **Class I**, meaning a clear-anodize thickness of about:
A. 0.5–1 µm   B. ≥18 µm (0.7 mil)   C. 1–2 mm   D. There is no thickness requirement

Q16. (Short answer) Name the **three** common seal methods and give **one tradeoff** of each. On a two-step part, what does sealing lock in?

Q17. (Short answer) [SAFETY GATE] List **three** pieces of PPE required at the sulfuric anodize / metal-salt color tanks, and name the special first-aid concern if your desmut or cold seal contains **fluoride.**

Q18. Sealing **bloom** (a chalky film) on a dark-bronze part is usually caused by:
A. Too little AC voltage during coloring   B. The part being too clean   C. Using DI water   D. Hard water, wrong pH, or an over-concentrated/contaminated seal

Q19. (Short answer) [SAFETY GATE] Name the **flammable gas** produced at the cathode during the Step-1 anodize and **two** controls that keep it from becoming a fire hazard.

Q20. (Short answer) Briefly contrast the **three ways to color anodized aluminum** (dye, integral, two-step) — one defining trait of each.

> ⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**
> Questions 3, 9, 13, 17, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

⚠ **HEADS-UP**

Keep this page out of the trainee's copy. Questions **3, 9, 13, 17, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Q1. C — Color is **fine metal (tin/nickel/cobalt) deposited deep in the pore bottoms with AC**, after a clear anodize.

Q2. A — The part is the **anode**, and oxide is grown out of the part itself.

Q3. D — **AC electrical current in an acidic metal-salt bath**, plus **toxic tin/nickel/cobalt salts** (Ni/Co sensitizers/carcinogen concern by inhalation) and their mist.

Q4. B — Darker = **more coloring time and/or higher AC voltage** (more deposited metal) — *not* a thicker anodize.

Q5. C — (Step 1) a **clear sulfuric (Type II) anodize**, then (Step 2) **AC electrolytic coloring** in a metal-salt bath.

Q6. In plating the part is the **cathode** and a **different metal is deposited onto it.** In anodizing the part is the **anode**, and the clear coating is **aluminum oxide grown out of the part's own surface** (oxygen from the bath + the part's aluminum) — nothing is deposited from elsewhere in Step 1.

Q7. A — **Tin (stannous sulfate)** — widest range, stable, best blacks, lowest toxicity of the three.

Q8. B — **More energy-efficient and far more consistent/repeatable** (color decoupled from coating thickness and alloy), with **excellent blacks.**

Q9. The **caustic etch** (high pH / sodium hydroxide) and the **sulfuric anodize/desmut/color baths** (low pH / acid) — both cause severe burns. Rule: **always add ACID to WATER, never water to acid.**

Q10. D — **AC (alternating current)** — it exploits the barrier layer's rectifying behavior to deposit metal at the pore base.

Q11. C — The **Step-2 coloring time and AC voltage** (the clear-coat thickness is held to spec).

Q12. Smut is the dark film of alloying elements (copper, silicon, iron) **left behind by the caustic etch**; removed at the **desmut/deox** station (acid dip). A uniform, well-grown, smut-free clear coat is the **foundation of uniform color** — an uneven or smutty clear coat colors **blotchy**, and the color tank can't fix it.

Q13. C — **Nickel and cobalt are skin sensitizers and carcinogen concerns by inhalation** — control the mist (local exhaust), keep off skin, no hand-to-mouth.

Q14. A — **Catch it before sealing and return it to the color tank** for more time/voltage to reach the standard; if already sealed, **strip and re-anodize.** (You can't darken with dye or paint.)

Q15. B — **≥18 μm (0.7 mil)** — architectural Class I.

Q16. Hot DI water (~96–100 °C) — simple/metal-free but high energy and bloom-prone; **nickel-acetate (~80–90 °C)** — efficient/excellent retention but contains nickel; **cold nickel-fluoride (~25–30 °C)** — fast/low-energy but nickel + fluoride hazard and tight control. Sealing locks in **both the corrosion resistance AND the deposited color metal** (closes the pores around it).

Q17. PPE (any three): sealed chemical splash goggles, face shield, chemical-resistant gloves, apron/suit, chemical-resistant boots. **Fluoride concern:** fluoride causes **deep, delayed, systemically toxic burns** — requires specific first-aid (e.g., **calcium gluconate** per SDS/medical plan) and immediate medical attention; ordinary flushing alone may not be enough.

Q18. D — **Hard water, wrong pH, or an over-concentrated/contaminated seal.** (DI water and correct pH/concentration prevent it.)

Q19. Hydrogen gas (at the Step-1 cathode). Controls (any two): **ventilation/exhaust running**, **no open flames/sparks/grinding nearby**, no smoking, keep ignition sources away.

Q20. Dye = organic color *in the pores* — cheap, any color, least lightfast (interior). **Integral** = color made *by the anodizing itself* (alloy + oxide) — durable but energy-intensive and alloy-sensitive, limited bronzes/black. **Two-step** = *metal deposited in the pore bottoms with AC* — durable, lightfast, consistent, energy-efficient — the modern architectural standard with the best blacks.

Answer-key letter distribution (MC): A: Q2, Q7, Q14 · B: Q4, Q8, Q15 · C: Q1, Q5, Q11, Q13 · D: Q3, Q10, Q18 — a healthy spread across all four letters (A 3 · B 3 · C 4 · D 3 across the 13 multiple-choice items; the other seven are short-answer).

★ **REMEMBER**

A score of 16/20 passes — but every safety question (3, 9, 13, 17, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people.

PLATING POSTERS

PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

CERTIFICATE OF COMPLETION

TWO-STEP ELECTROLYTIC COLOR ANODIZING TRAINING PROGRAM

This certifies that

____

OPERATOR NAME

has successfully completed the **Two-Step Electrolytic Color Anodizing** training course — covering the full two-step workflow (Inspection & Racking, Clean, Etch, Rinse, Desmut/Deox, **Clear Sulfuric Anodize — Step 1**, Rinse, **AC Electrolytic Coloring — Step 2**, Rinse, Seal, Rinse/Dry, and Final Inspection), the fundamental concept that **the part is the anode and the clear oxide is grown out of the part itself**, the principle that **color is fine metal deposited deep in the pore bottoms using AC — its depth set by coloring time and voltage, not by coating thickness**, the three coloring routes (dye, integral, two-step) and why two-step is the modern architectural standard, dimensional growth, Class I clear-coat control, electrical-contact and racking practice, throwing-power and color/ΔE control, sealing chemistry and seal-quality testing, and — above all — the **safety practices** required of every operator (sulfuric-acid and caustic-etch burn control, hot-seal scald control, hydrogen and **DC + AC** electrical controls, **tin/nickel/cobalt metal-salt handling and mist control**, fluoride first-aid where applicable, spill response, and metal-salt waste segregation) — and has passed the Completion Test with a score of __________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ____________________ **Re-certification due:** ____________________

____

TRAINEE SIGNATURE

____

CERTIFYING TRAINER

____

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer/architectural specifications (e.g., AAMA 611, ASTM B580/B137/B244), and applicable regulatory requirements before production use. Sulfuric acid and caustic etch cause severe burns; seal tanks run near boiling; the color tank combines AC current with an acidic tin/nickel/cobalt metal-salt solution — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.

PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES · TWO-STEP ELECTROLYTIC COLOR ANODIZING TRAINING MANUAL